Social Media Business Intelligence

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1 Social Media Sentiment BI Analytics

1.1 Data Sources

Dataset: DMO Social Media Engagement Dataset

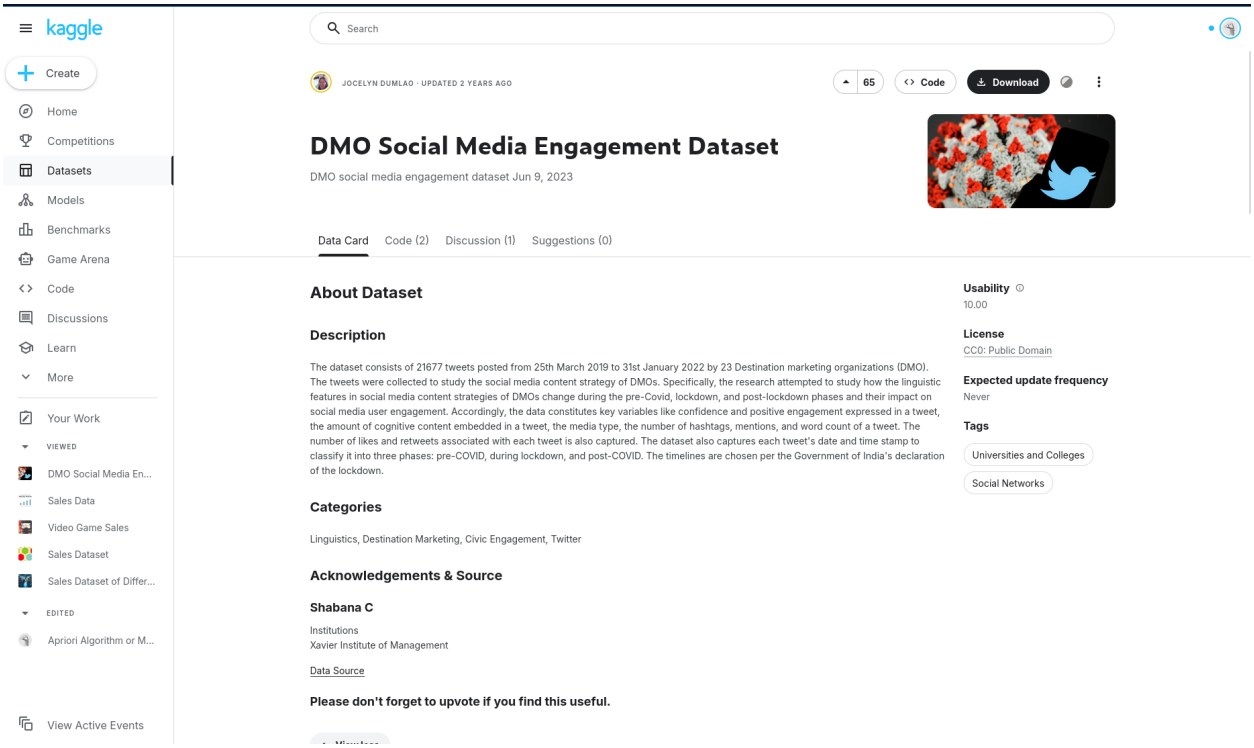


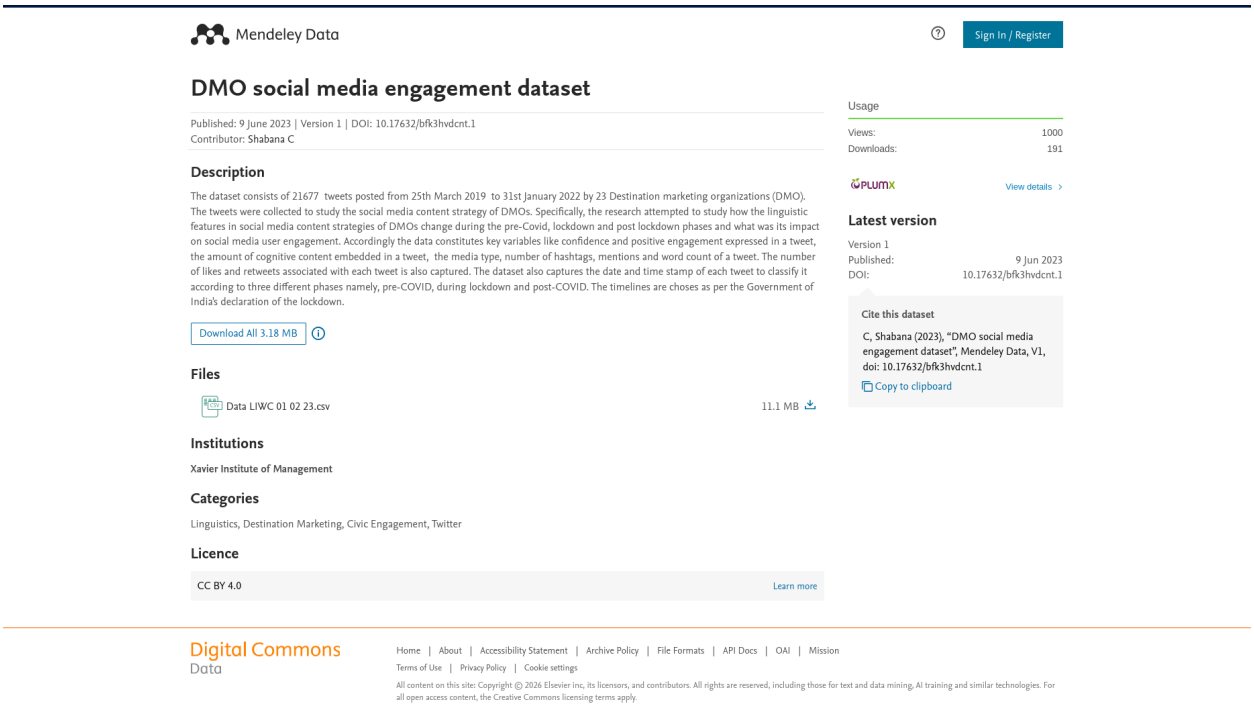
Figure 1: Kaggle DMO Social Media Engagement Dataset

Primary Source: DMO Social Media Engagement Dataset

This dataset contains 21,677 tweets collected from 23 Destination Marketing Organizations (DMOs) between March 25, 2019 and January 31, 2022. It was created to study how social media content strategies and linguistic features affect user engagement on Twitter.

2 Problem Statement

Social media platforms such as Twitter have become essential tools for organizations to communicate with their audiences, promote services, and build brand visibility. Destination Marketing



Mendeley Data

Sign In / Register

DMO social media engagement dataset

Published: 9 June 2023 | Version 1 | DOI: 10.17632/bfk3hvdcnt.1
Contributor: Shabana C

Description

The dataset consists of 21677 tweets posted from 25th March 2019 to 31st January 2022 by 23 Destination marketing organizations (DMO). The tweets were collected to study the social media content strategy of DMOs. Specifically, the research attempted to study how the linguistic features in social media content strategies of DMOs change during the pre-Covid, lockdown and post lockdown phases and what was its impact on social media user engagement. Accordingly the data constitutes key variables like confidence and positive engagement expressed in a tweet, the amount of cognitive content embedded in a tweet, the media type, number of hashtags, mentions and word count of a tweet. The number of likes and retweets associated with each tweet is also captured. The dataset also captures the date and time stamp of each tweet to classify it according to three different phases namely, pre-COVID, during lockdown and post-COVID. The timelines are choses as per the Government of India's declaration of the lockdown.

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Files

Data LIWC 01 02 23.csv 11.1 MB

Institutions

Xavier Institute of Management

Categories

Linguistics, Destination Marketing, Civic Engagement, Twitter

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C, Shabana (2023), "DMO social media engagement dataset", Mendeley Data, V1, doi: 10.17632/bfk3hvdcnt.1

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Figure 2: Mendley DMO Social Media Engagement Dataset

Organizations (DMOs), in particular, rely heavily on Twitter to engage potential tourists, share promotional content, and maintain public interest in travel destinations. Engagement on these platforms measured through likes, replies, and retweets serves as a key indicator of how effectively content resonates with users.

Despite the availability of large volumes of social media data, it remains unclear which specific factors drive higher engagement. Tweet performance can be influenced by multiple variables, including content type (e.g., image, video, or text), posting time, emotional tone, linguistic features, and the size or influence of the account posting the content. Additionally, external conditions such as different time periods (e.g., pre-COVID, lockdown, and post-lockdown phases) may also impact how audiences interact with content. Understanding these factors is crucial for organizations seeking to optimize their communication strategies and maximize audience reach.

This project addresses the challenge of identifying the key drivers of social media engagement by analyzing a dataset of tweets collected from Destination Marketing Organizations. The dataset contains information on engagement metrics, sentiment scores, linguistic characteristics, posting

time, and audience attributes such as follower count. However, the raw dataset contains inconsistencies such as redundant columns, unclear naming conventions, and mixed data types, which must be addressed before meaningful analysis can be performed.

The objective of this project is therefore twofold: first, to clean, transform, and model the dataset into a structured format suitable for analysis; and second, to explore and identify patterns that explain how different factors influence tweet engagement. By applying data modeling techniques and analytical methods, the project aims to uncover insights into how content strategy, timing, sentiment, and audience characteristics affect user interaction on social media.

The findings from this analysis can support organizations in making data-driven decisions about their social media strategies. Specifically, it can help DMOs determine the most effective types of content to post, the optimal times for posting, and how to tailor messaging to maximize engagement. Ultimately, the project contributes to a better understanding of how social media communication strategies can be optimized to improve audience engagement and overall digital marketing effectiveness

3 Data Dictionary

This provides a detailed description of all variables used in the Social Media Engagement dataset obtained from Kaggle. The purpose of this data dictionary is to enhance data understanding, ensure consistency and support effective analysis.

The dataset captures social media performance using engagement metrics, temporal variables, audience characteristics, content attributes, sentiment scores and linguistic features derived from LIWC analysis.

While variables such as likes, retweets, replies and follower counts quantify observable user interactions, they do not fully explain the underlying drivers of engagement. To address this gap the dataset incorporates LIWC based variables that model the psychological, emotional and communicative properties of text.

These models allow analysts to move beyond surface level metrics and examine how language it-

	Column Name	Business Meaning	Data Type	Category
	Valid 100% Error 0% Empty 0% 30 distinct, 30 unique	Valid 100% Error 0% Empty 0% 30 distinct, 30 unique	Valid 53% Error 0% Empty 47% 5 distinct, 2 unique	Valid 50% Error 0% Empty 50% 7 distinct, 3 unique
1	Retweet Count	Number of times post was reshared	Whole Number	Engagement
2	Reply Count	Number of replies	Whole Number	Engagement
3	Like Count	Number of likes	Whole Number	Engagement
4	Qoute Count	Number of quote shares	Whole Number	Engagement
5	Engagement	Total interactions on post	Whole Number	Engagement
6	Date	Date post was published	Date	Time
7	Open Hours	Hour posted	Whole Number	Time
8	Day	Day of week	Text	Time
9	Day Type	Weekday or weekend	Text	Time
10	4 Phase	Campaign phase classification	Text	Campaign
11	Number of followers	Followers at time of post	Whole number	Audience
12	State	Location of audience	Text	Demographic
13	Post Type	Format	Text	Content
14	Content Type	Topic Category	Text	Content
15	Word Count	Number of words in post	Whole Number	Content

Figure 3: alt text

16	Clout	Confidence and leadership tone score	Text	
17	Cognition	Degree of analytical thinking words		
18	Affect	Overall emotional expression level		
19	Emotion	Total emotional language used		
20	Positive Emotion	Positive feeling words percentage		
21	Negative Emotion	Negative feeling words percentage		
22	We Percentage	Percentage of collective pronouns		
23	I Percentage	Percentage of self-focused pronouns		
24	They Percentage	Percentage of third person pronouns		
25	Tentativity	Level of uncertainity words used		
26	Motivations	Words related to goals and drives		
27	Insights	Words reflecting realization or thinking		
28	Causation	Cause effect language used		
29	Conflict	Presence of disagreement		
30	Certainty	Confidence or absolute language		

Figure 4: alt text

self shapes audience perception, interaction and behavioral response. The data dictionary therefore plays a critical role in translating these complex variables into meaningful interpretations that can be applied within Power BI for reporting and decision making. The psychological process model captures the emotional and cognitive dimensions of language, focusing on how individuals express feelings and thoughts within social media content. Variables such as affect, emotion, positive emotion and negative emotion quantify the intensity and polarity of emotional expression, enabling analysts to distinguish between content that evokes positive reactions, such as 3, 2 and 1 and content that conveys negativity, such as -3, -2 or -1. Cognition extends this analysis by measuring the degree of analytical thinking present in the text, reflecting the use of reasoning, problem solving and intellectual framing.This model is particularly valuable in understanding the balance between emotional appeal and rational communication, as highly emotional posts may drive immediate engagement, while cognitively rich posts may foster deeper audience reflection and long term value. By interpreting these variables collectively, analysts can identify whether emotional resonance or intellectual depth is more effective in influencing engagement outcomes.

The social language model examines how language reflects interpersonal relationships and social orientation, primarily through the use of pronouns. Variables such as we percentage, I percentage and they percentage provide insight into whether content is framed collectively, personally or externally. A higher use of (we) indicates inclusive and community oriented messaging, which can strengthen audience connection and foster a sense of belonging. A higher (I) percentage reflects a personal or individualistic tone, often associated with storytelling, personal experiences or influencer driven content. The use of (they) suggests a focus on external groups or third parties which may introduce elements of comparison, distinction or even criticism. This model is essential for understanding how linguistic framing influences audience engagement as different social orientations can evoke varying emotional and behavioral responses. For instance, inclusive language may enhance trust and loyalty while personal narratives may increase relatability and authenticity.

The cognitive and reasoning model explores how ideas are structured, communicated and interpreted within text, emphasizing the role of logical and reflective language. Variables such as insights, causation, tentativity and certainty capture distinct aspects of thinking and reasoning. Insight related words indicate moments of realization or understanding, suggesting that the content

encourages reflection or learning. Causation measures the presence of cause and effect relationships, highlighting whether the text provides explanations or logical arguments. Tentativity reflects uncertainty or openness, often expressed through words like 'maybe' or 'perhaps' while certainty conveys confidence and decisiveness through definitive language. This model provides a understanding of how communication style influences audience perception, as content that balances clarity with openness may be perceived as both credible and engaging. Us analysts can use these variables to determine whether audiences respond more positively to assertive, confident messaging or to exploratory, thought provoking content.

The motivation and influence model focuses on the expression of goals, intentions and persuasive power within language, capturing how content inspires action and conveys authority. The motivations variable identifies language related to desires, ambitions and goal oriented thinking, reflecting the extent to which content encourages forward looking behavior or aspiration. Clout, on the other hand, measures the level of confidence, leadership, and influence embedded in the text. High clout scores indicate authoritative and self assured communication, which can enhance credibility and persuasion, particularly in marketing or leadership contexts. This model is critical for understanding how language drives behavioral outcomes, as motivational and high-clout messaging is often associated with increased engagement, conversions and trusting the audience. By analyzing these variables, organizations can refine their communication strategies to emphasize empowerment, confidence and clear calls to action.

The conflict and interaction model captures the presence of tension, disagreement or opposition within content, providing insight into how controversial or confrontational language influences engagement. The conflict variable identifies words and expressions associated with argument, criticism or debate highlighting whether the content introduces opposing viewpoints or challenges existing perspectives. Conflict driven content can attract attention and stimulate interaction, it also carries potential risks, including negative sentiment, audience polarization and reputational damage. This model is particularly relevant in the context of social media, where provocative or controversial posts often generate high levels of engagement but may not always align with brand objectives. By incorporating this variable into analysis, analysts can evaluate the trade off between visibility and sentiment, helping organizations make informed decisions about the tone

and positioning of their content.

The sentiment model provides a consolidated measure of the overall emotional tone of content, simplifying complex linguistic patterns into interpretable scores. Variables such as positive sentiment, negative sentiment, and total sentiment summarize the emotional direction and intensity of text, offering a high-level view of how content is likely to be perceived by audiences.The more granular emotional variables in the psychological process model, sentiment scores aggregate multiple dimensions of language into a single metric, making them particularly useful for comparative analysis and dashboard visualization in Power BI. Positive sentiment typically indicates favorable reception and alignment with audience expectations, while negative sentiment may signal dissatisfaction or controversy. Total sentiment provides a net measure, balancing positive and negative influences to give an overall assessment of tone. This model is essential for quickly identifying trends, benchmarking performance and guiding strategic decisions related to content creation and audience engagement. THANK YOU!

4 ETL & Power Query

4.1 Extract

The raw data had 41 columns with redundancy present. Certain columns represented roughly the same thing and were not needed for analysis.

Sno	X	State	conversation_id	retweet_count	reply_count	like_count	quote_count	Buzz	id	Date1	OpnHours	DateDay	Day	Time	3-Phase	4-phase	Followers	Status text
																		World Tourism Day, Tourism for inclus
1	1	Assam	S1438213588544409600	0	1	1	0	3	S1438213588544409600	15-09-2021	Non-working Hours	Wednesday	weekday	Non-buisness hours	Post Lockdown	Post Lockdown	5164	In conjunction with World Tourism Day
2	2	Assam	S1438437895270846464	0	1	1	0	3	S1438437895270846464	16-09-2021	Working Hours	Thursday	weekday	Business hours	Post Lockdown	Post Lockdown	5164	In tune with this year's World Tourism

Figure 5: Raw Data Screenshot 1

Vividness	ContentType	WC	Clout	Cognition	Affect	emotion	emo_pos	emo_neg	we	tentat	Drives	i	we	they	insight	cause	discrep	certitude	Positive	Negative	Total_Sentiment
photo	Information	31	40.06	3.23	3.23	3.23	0	3.23	0	0	3.23	0	0	0	3.23	0	0	0	1	-2	1
photo	Information	44	40.06	0	2.27	2.27	0	2.27	0	0	0	0	0	0	0	0	0	0	1	-1	0

Figure 6: Raw Data Screenshot 2

4.2 Transform

4.2.1 Data Cleaning

4.2.1.1 Dropped Columns

The `Sno` and `X` columns are roughly the same thing. `Sno` handled a continuous count throughout the whole dataset while `X` reset to 1 every time the `State` changed. Moreover, they are not needed thus they were dropped from the dataset.

The `conversation_id` and `id` field are pretty much the same values and are not needed thus they are dropped. The `Status text` field is dropped since sentiment analysis has already been performed on it and hence has become obsolete.

The `3-Phase` and `4-Phase` columns represent roughly the same concepts with `4-Phase` encompassing more categorical splitting of time. Hence, only one is needed and the more descriptive one (`4-Phase`) is retained.

The columns `Time` and `OpnHrs` are the same values in practice hence since `OpnHrs` is a better descriptor it is the one that is retained. The Linguistic Inquiry and Word Count (LWIC) columns except `Positive`, `Negative`, `Total_Sentiment`, and `WC` are dropped since they offer no explaining power for business problems. They are used as intermediate columns to compute the positive, negative, and total sentiments from a piece of content.

4.2.1.2 Renamed Columns

The `Date1` column is renamed to `Date` and `DateDay` to `Day`. The `Day` column is renamed to `Day_Type`. The column `OpnHrs` is renamed to `Open_Hours` to align with good grammar. The `Followers` column is renamed to `Number_Of_Followers` for increased clarity. The `Vividness` column is renamed to `Post_Type` for it to be a lot more functional- since it holds the different media types a post can be. The `WC` column is expanded to `Word_Count`.

While renaming the columns the snake and camel cases were merged thus following the pattern capitalizing the first letter and inserting an underscore between words (`Retweet_Count` for example).

4.2.1.3 Computed Columns

Using the `Number_Of_Followers` column the `Influencer_Status` column was created. The following bands were employed.

- <= 10,000 → Nano Influencer
- <= 100,000 → Micro Influencer
- <= 1,000,000 → Macro Influencer
- > 1,000,000 → Celebrity

From these bands, in our entire dataset the Celebrity status was the third highest one. And the Nano Influencer status was the least had the lowest representation.

Using the `Word Count` column the `Verbosity` column was created with the following categories.

- <= 20 → Low
- <= 60 → Medium
- <= 120 → High
- > 120 → Extreme

The Medium band holds the largest number of posts.

4.2.1.4 Remapped Columns

The `State` and `Open_Hours` columns had inconsistent naming and abbreviation of values thus they were remapped.

For `State` the map was:

And for `Open_Hours`:

4.2.1.5 Data Type Conversion

Column Name	Data Type
State	Text
Retweet_Count	Whole Number
Reply_Count	Whole Number
Like_Count	Whole Number
Quote_Count	Whole Number

Column Name	Data Type
Engagement	Whole Number
Date	Date
Open_Hours	Text
Day	Text
Day_Type	Text
4_Phase	Text
Number_Of_Followers	Whole Number
Post_Type	Text
Content_Type	Text
Word_Count	Whole Number
Positive_Sentiment	Whole Number
Negative_Sentiment	Whole Number
Total_Sentiment	Whole Number
Influencer_Status	Text
Verbosity	Text

4.3 Load

4.3.1 Power Query

4.3.1.1 Fact Table Social Media Engagement

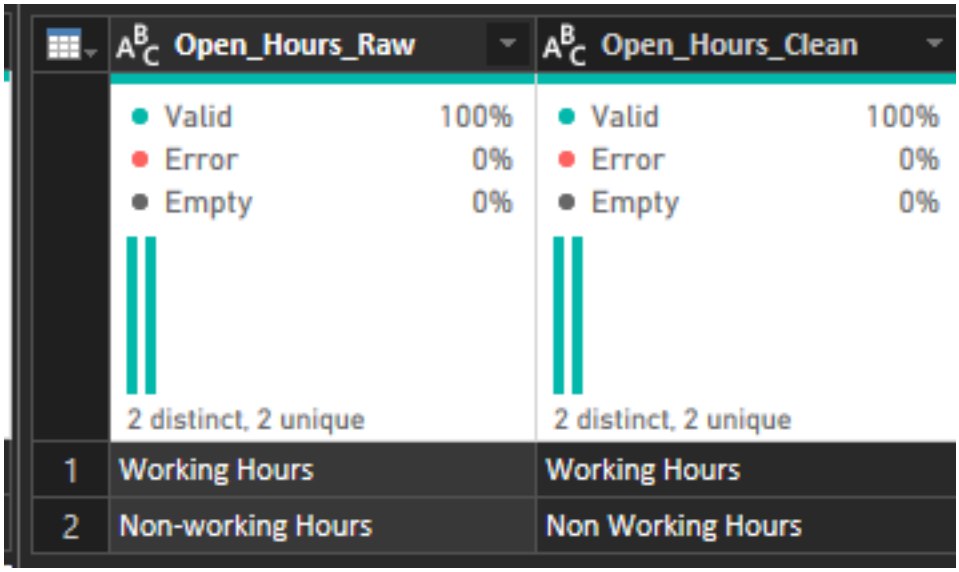
4.3.1.2 Dimension Table State

4.3.1.3 Dimension Table Post Type

4.3.1.4 Dimension Table Content Type

	StateRaw	StateClean
	Valid 100% Error 0% Empty 0% 23 distinct, 23 unique	Valid 100% Error 0% Empty 0% 23 distinct, 23 unique
1	Assam	Assam
2	Bihar	Bihar
3	Chattisgarh	Chattisgarh
4	Delhi	Delhi
5	Goa	Goa
6	Gujarat	Gujarat
7	HP	Himachal Pradesh
8	J and K	Jammu and Kashmir
9	Jharkhand	Jharkhand
10	Karnataka	Karnataka
11	Kerala	Kerala
12	Maharashtra	Maharashtra
13	Mizoram	Mizoram
14	MP	Madhya Pradesh
15	Nagaland	Nagaland
16	Odisha	Odisha
17	Rajasthan	Rajasthan
18	Tamilnadu	Tamilnadu
19	Telangana	Telangana
20	Tripura	Tripura
21	UP	Uttar Pradesh
22	Uttarkhand	Uttarkhand
23	WestBengal	West Bengal

Figure 7: State Map



	Open_Hours_Raw	Open_Hours_Clean
	Valid 100%	Valid 100%
	Error 0%	Error 0%
	Empty 0%	Empty 0%
	2 distinct, 2 unique	2 distinct, 2 unique
1	Working Hours	Working Hours
2	Non-working Hours	Non Working Hours

Figure 8: Open Hours Map

4.3.1.5 Dimension Table Phase

4.3.1.6 Dimension Table Open Hours

4.3.1.7 Dimension Table Date

4.3.1.8 Dimension Table Day Power Query

4.3.1.9 Dimension Table Day Type

4.3.1.10 Dimension Table Influencer Status

4.3.1.11 Dimension Table Verbosity

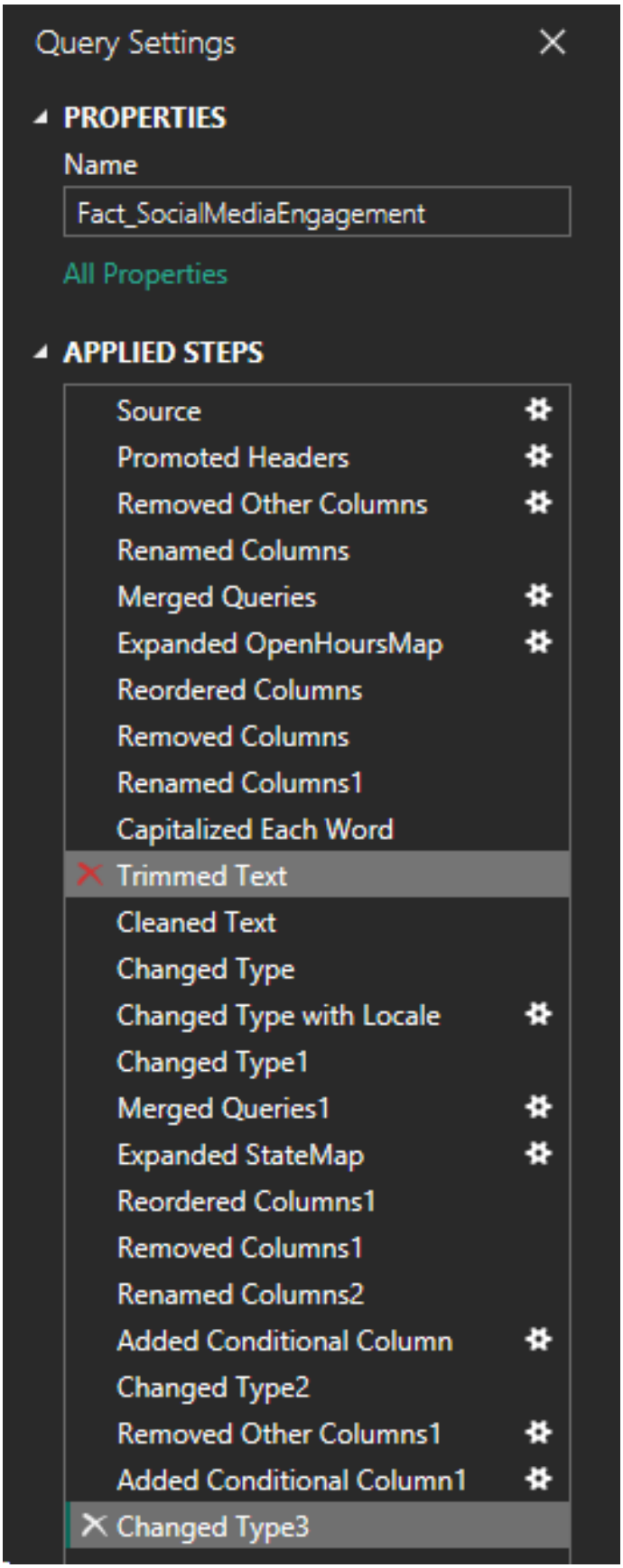


Figure 9: Fact Table Social Media Engagement Power Query

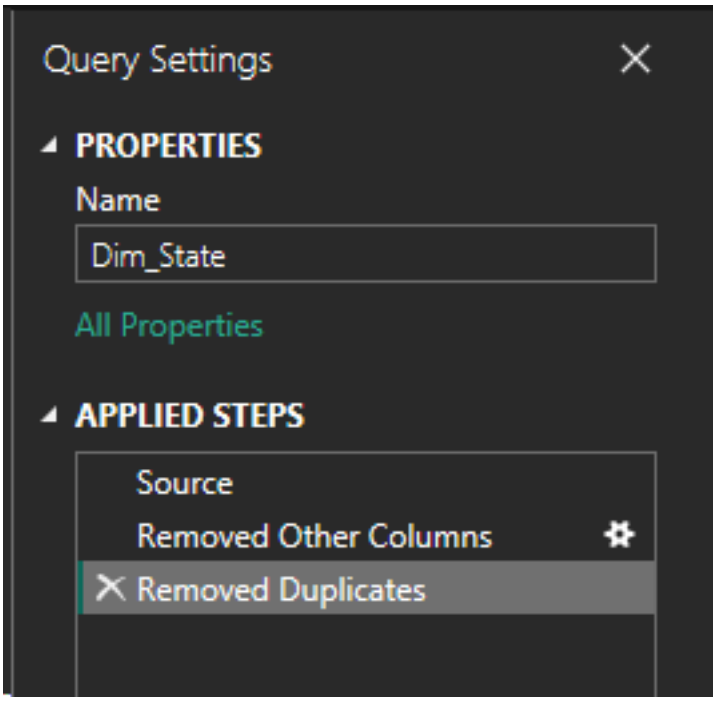


Figure 10: Dimension Table State Power Query

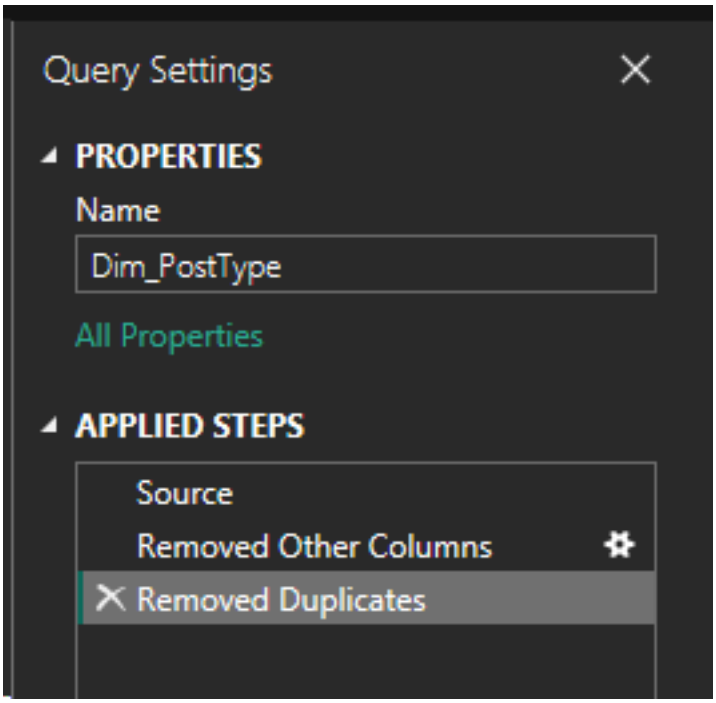


Figure 11: Dimension Table Post Type Power Query

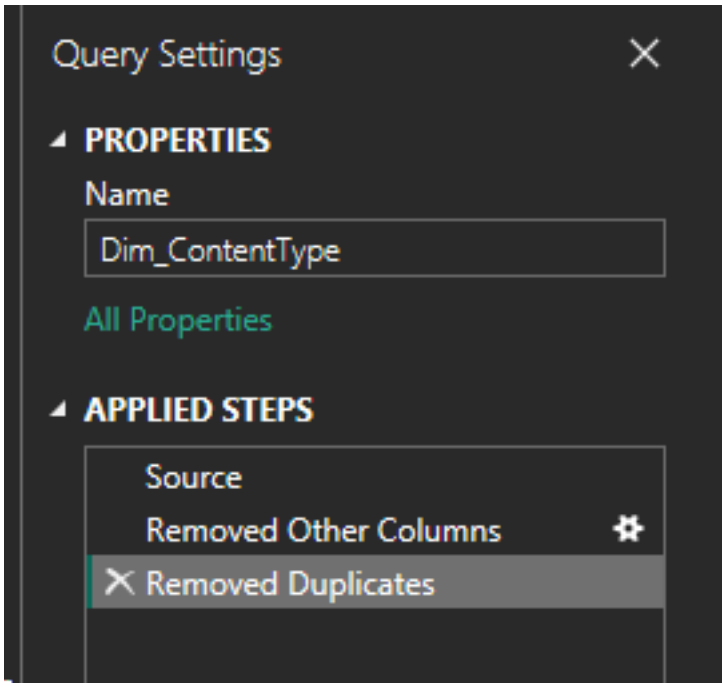


Figure 12: Dimension Table Content Type Power Query

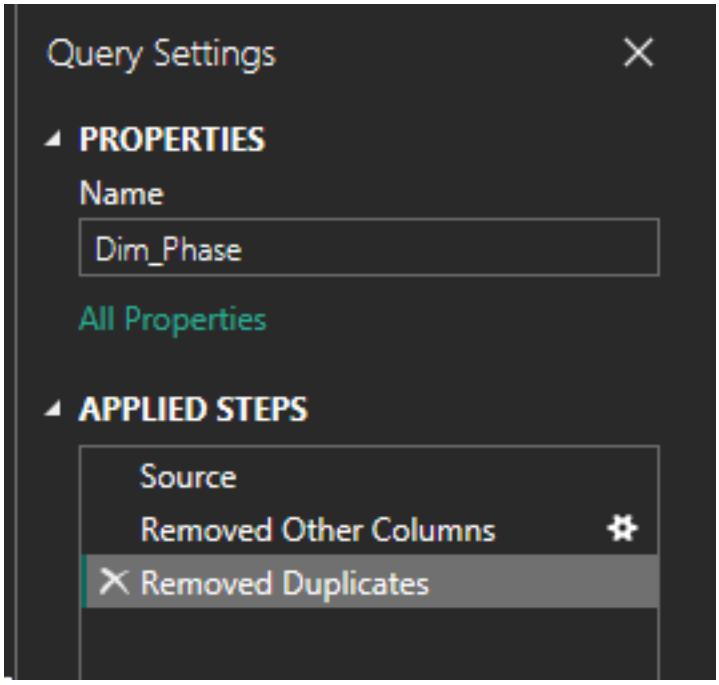


Figure 13: Dimension Table Phase Power Query

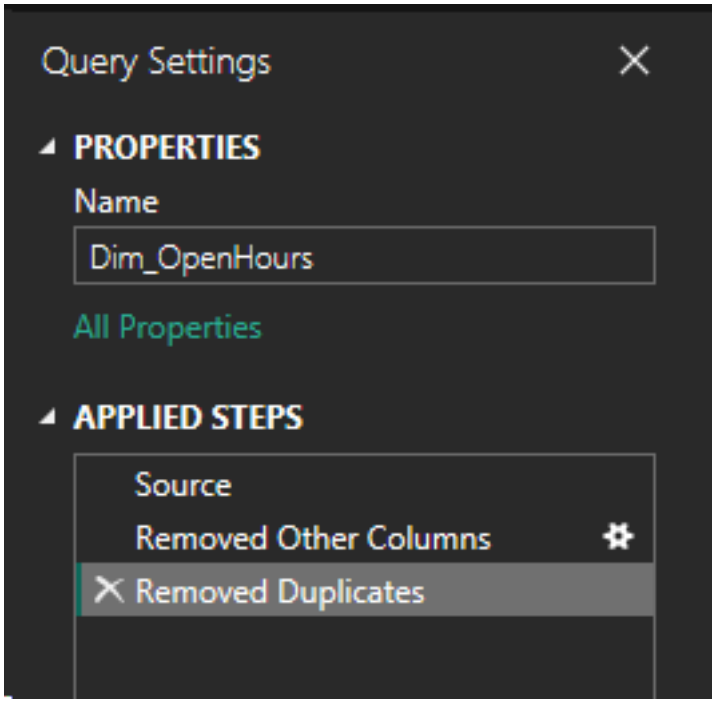


Figure 14: Dimension Table Open Hours Power Query

5 Data Architecture

5.1 Overview

The data model is designed using a star schema. This means there is one main table (fact table) in the center, and several smaller tables (dimension tables) around it.

- The fact table stores the main values (like engagement and likes)
- The dimension tables describe the data (like date, content type, state, etc.)

All the dimension tables connect to the fact table, which makes it easy to analyze the data from different angles.

5.2 Fact Table

5.2.1 Fact Table Social Media Engagement

This is the main table in the model.

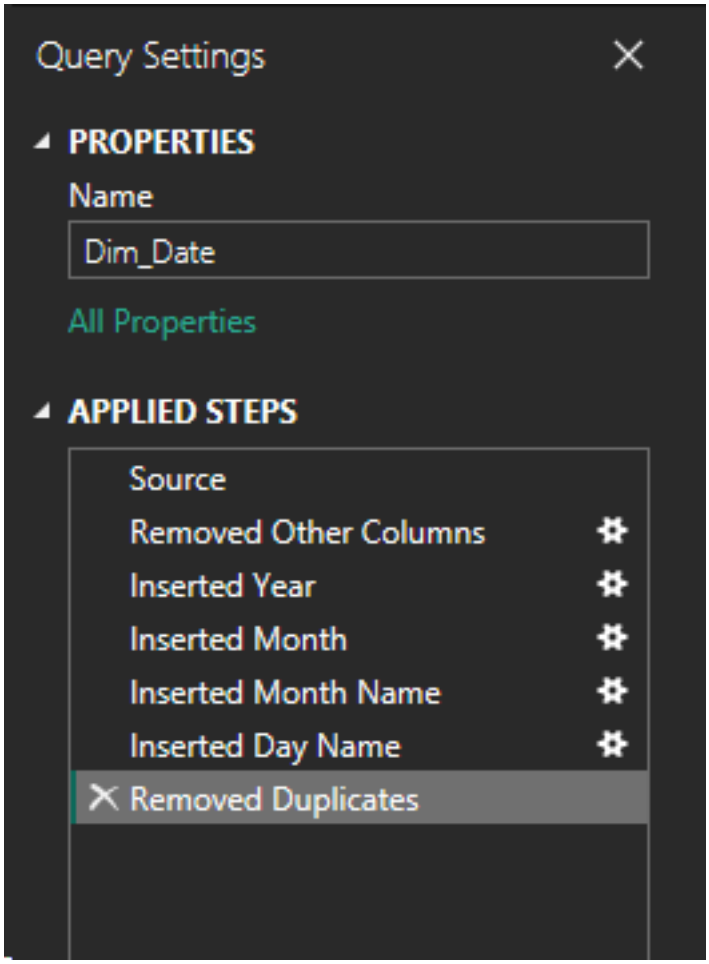


Figure 15: Dimension Table Date Power Query

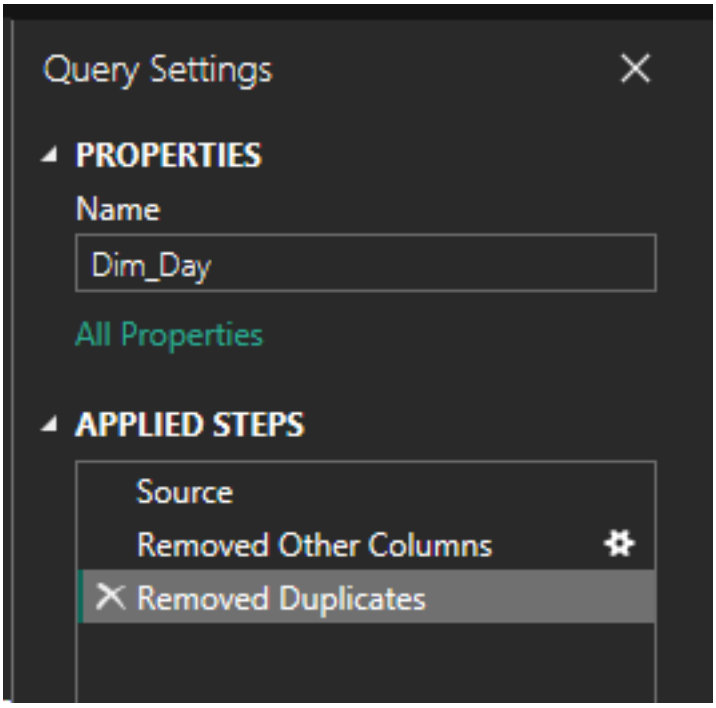


Figure 16: Dimension Table Day Power Query

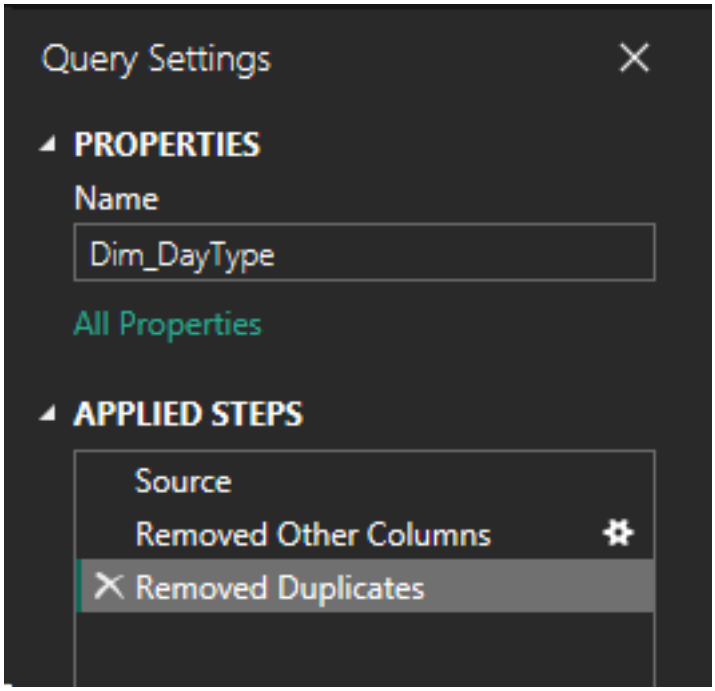


Figure 17: Dimension Table Day Type Power Query

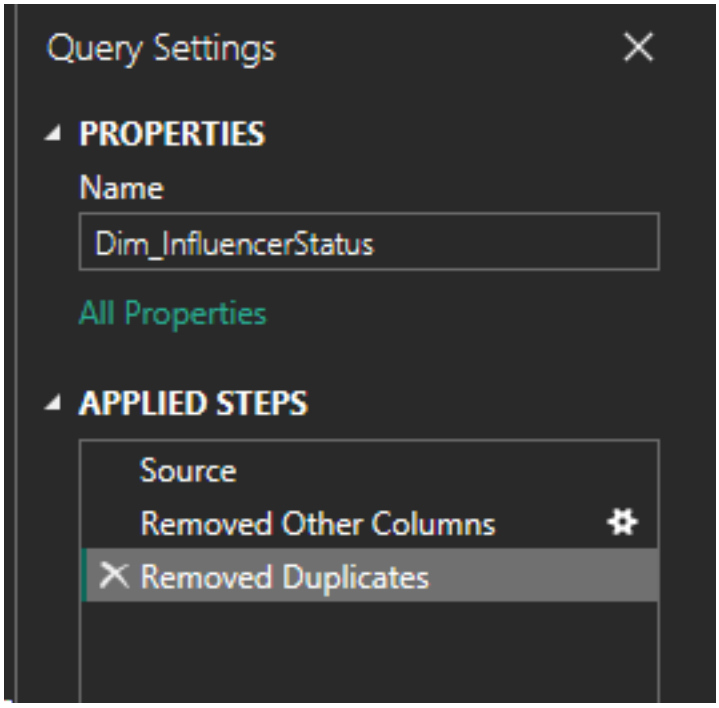


Figure 18: Dimension Table Influencer Status Power Query

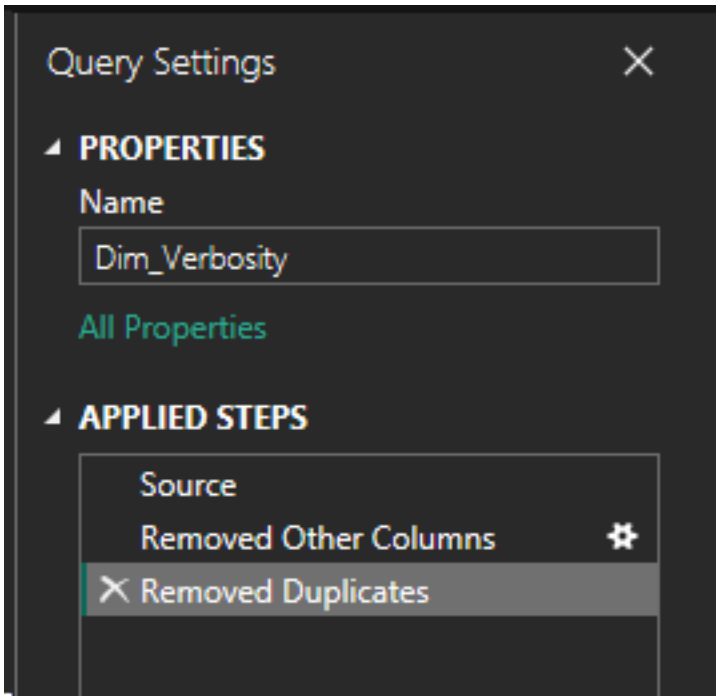


Figure 19: Dimension Table Verbosity Power Query

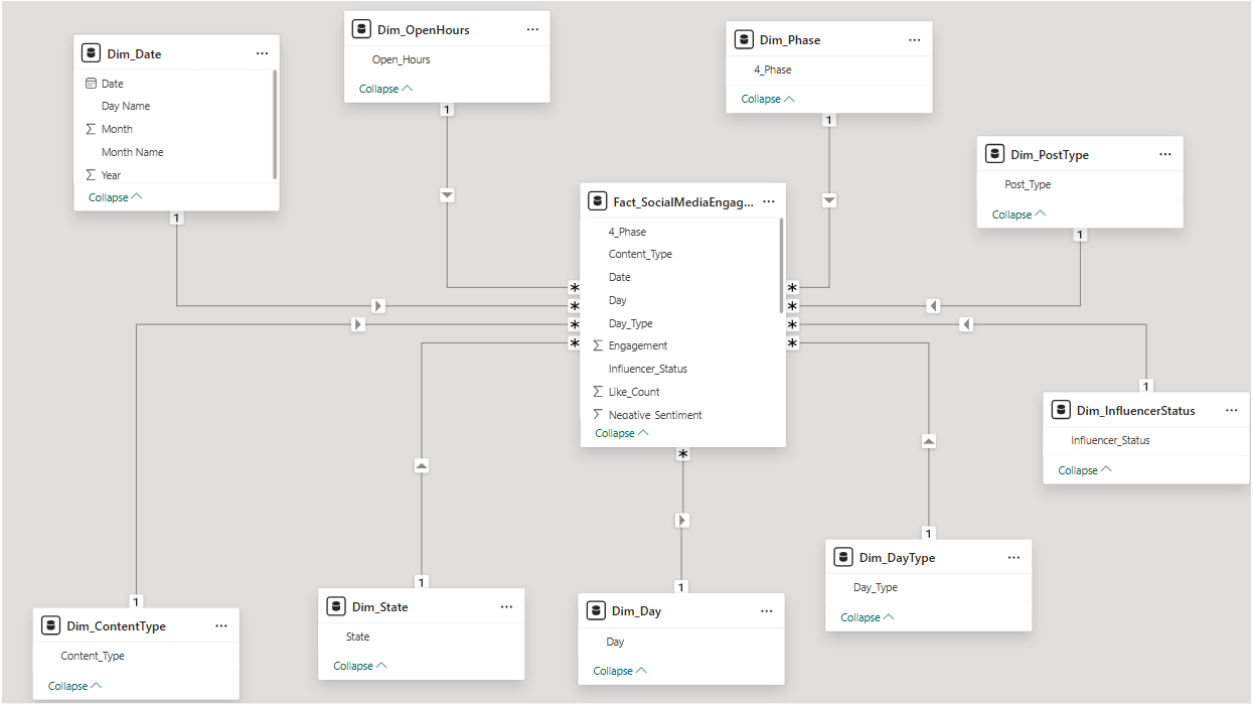
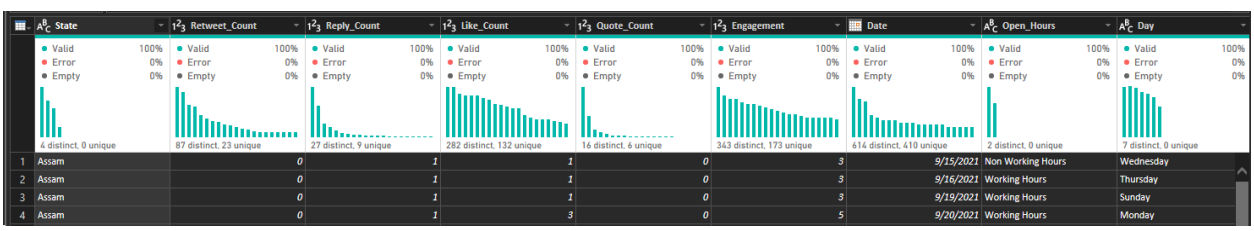


Figure 20: alt text

Each row represents one tweet.



	State	Retweet_Count	Reply_Count	Like_Count	Quote_Count	Engagement	Date	Open_Hours	Day
1	Assam	0	1	1	0	3	9/15/2021	Non Working Hours	Wednesday
2	Assam	0	1	1	0	3	9/16/2021	Working Hours	Thursday
3	Assam	0	1	1	0	3	9/19/2021	Working Hours	Sunday
4	Assam	0	1	3	0	5	9/20/2021	Working Hours	Monday

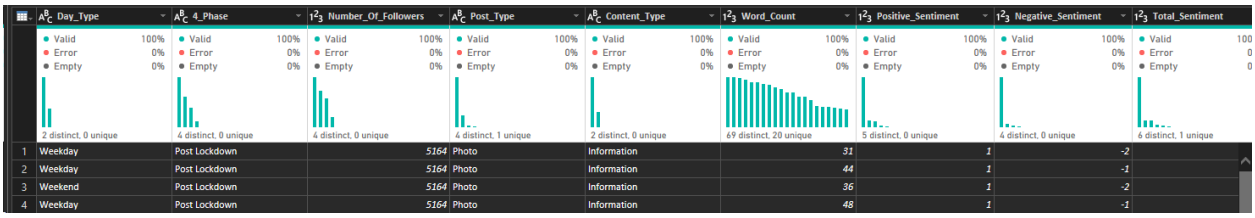
Figure 21: Fact Table Social Media Engagement Snippet 1

This table is used to:

- Calculate totals (e.g. total engagement)
- Compare values (e.g. engagement by content type)

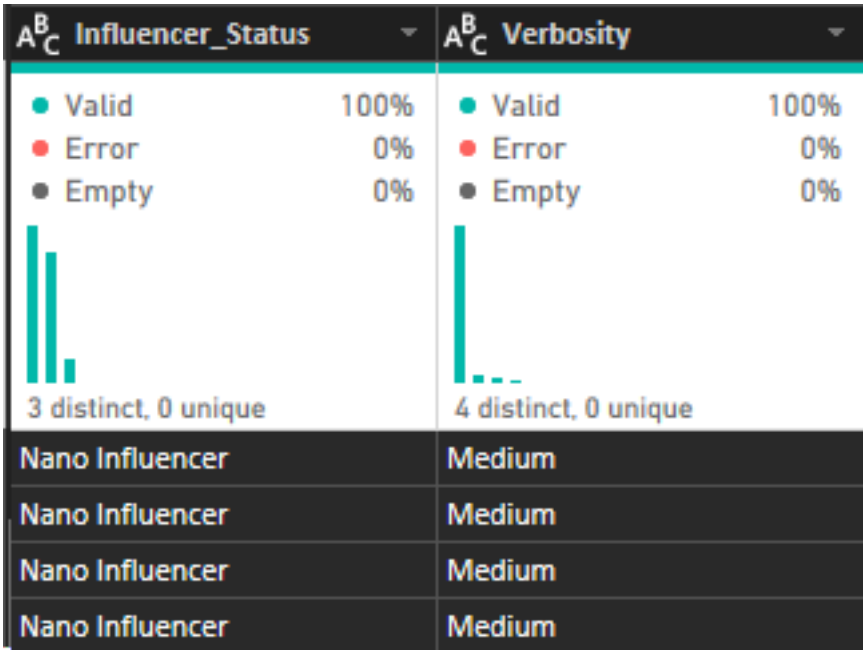
5.3 Dimension Tables

These tables give extra information about the data in the fact table.



	Day_Type	4_Phase	Number_Of_Followers	Post_Type	Content_Type	Word_Count	Positive_Sentiment	Negative_Sentiment	Total_Sentiment
	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100
	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0
	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0
	2 distinct, 0 unique	4 distinct, 0 unique	4 distinct, 0 unique	4 distinct, 1 unique	2 distinct, 0 unique	69 distinct, 20 unique	5 distinct, 0 unique	4 distinct, 0 unique	6 distinct, 1 unique
1	Weekday	Post Lockdown	5164	Photo	Information	31	1	-2	
2	Weekday	Post Lockdown	5164	Photo	Information	44	1	-1	
3	Weekend	Post Lockdown	5164	Photo	Information	36	1	-2	
4	Weekday	Post Lockdown	5164	Photo	Information	48	1	-1	

Figure 22: Fact Table Social Media Engagement Snippet 2



Influencer_Status	Verbosity
Valid 100%	Valid 100%
Error 0%	Error 0%
Empty 0%	Empty 0%
3 distinct, 0 unique	4 distinct, 0 unique
Nano Influencer	Medium
Nano Influencer	Medium
Nano Influencer	Medium
Nano Influencer	Medium

Figure 23: Fact Table Social Media Engagement Snippet 3

5.3.1 Dimension Table Date

	Date	Year	Month	Month Name	Day Name
	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%
	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%
	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%
	1000 distinct, 1000 unique	4 distinct, 0 unique	12 distinct, 0 unique	12 distinct, 0 unique	7 distinct, 0 unique
1	9/15/2021	2021	9	September	Wednesday
2	9/16/2021	2021	9	September	Thursday
3	9/19/2021	2021	9	September	Sunday
4	9/20/2021	2021	9	September	Monday
5	9/21/2021	2021	9	September	Tuesday
6	9/22/2021	2021	9	September	Wednesday
7	9/23/2021	2021	9	September	Thursday
8	9/25/2021	2021	9	September	Saturday

Figure 24: Dimension Table Date

Used to:

- Analyze trends over time
- Compare engagement by month or year

5.3.2 Dimension Table Day Power Query

Used to:

- Analyze engagement per day

5.3.3 Dimension Table Day Type

Used to:

- Compare weekend vs weekday performance

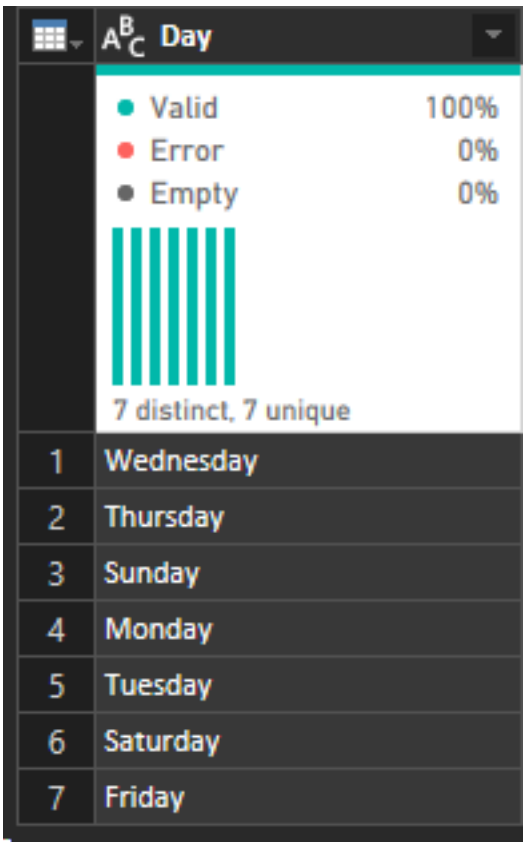


Figure 25: Dimension Table Day Power Query

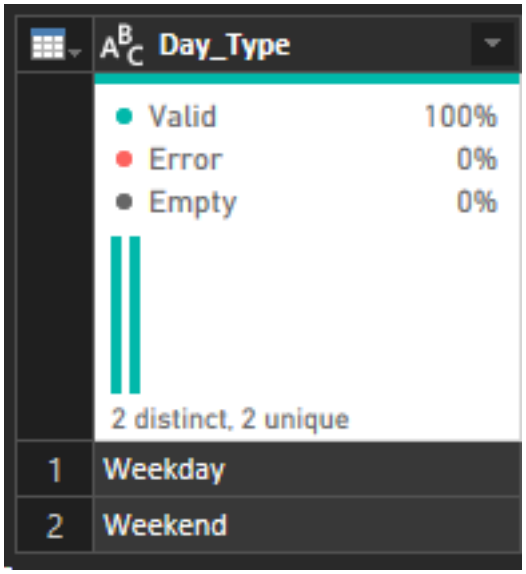


Figure 26: Dimension Table Day Type

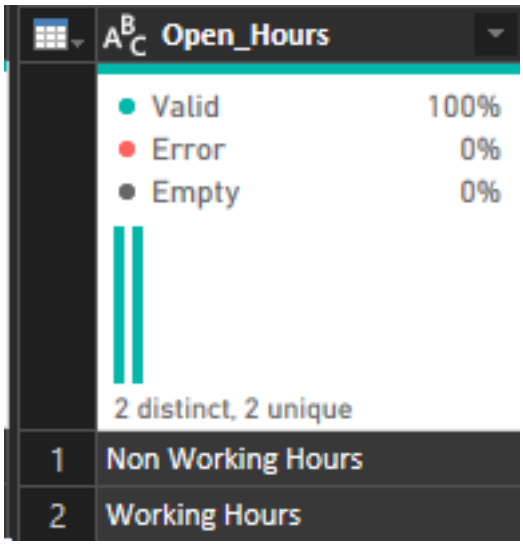


Figure 27: Dimension Table Open Hours

5.3.4 Dimension Table Open Hours

Used to:

- Find the best time to post

5.3.5 Dimension Table Content Type

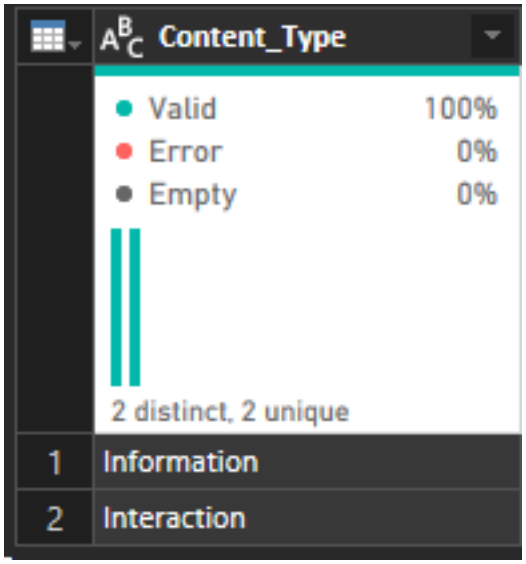


Figure 28: Dimension Table Content Type

Used to:

- See which content performs best

5.3.6 Dimension Table Post Type

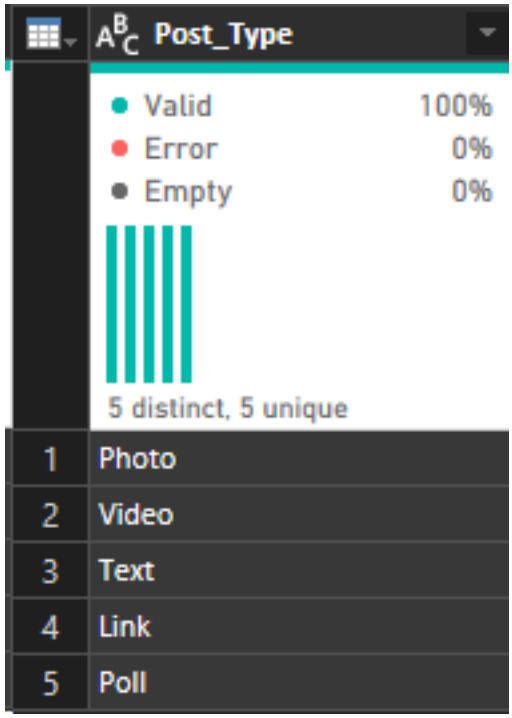


Figure 29: Dimension Table Post Type

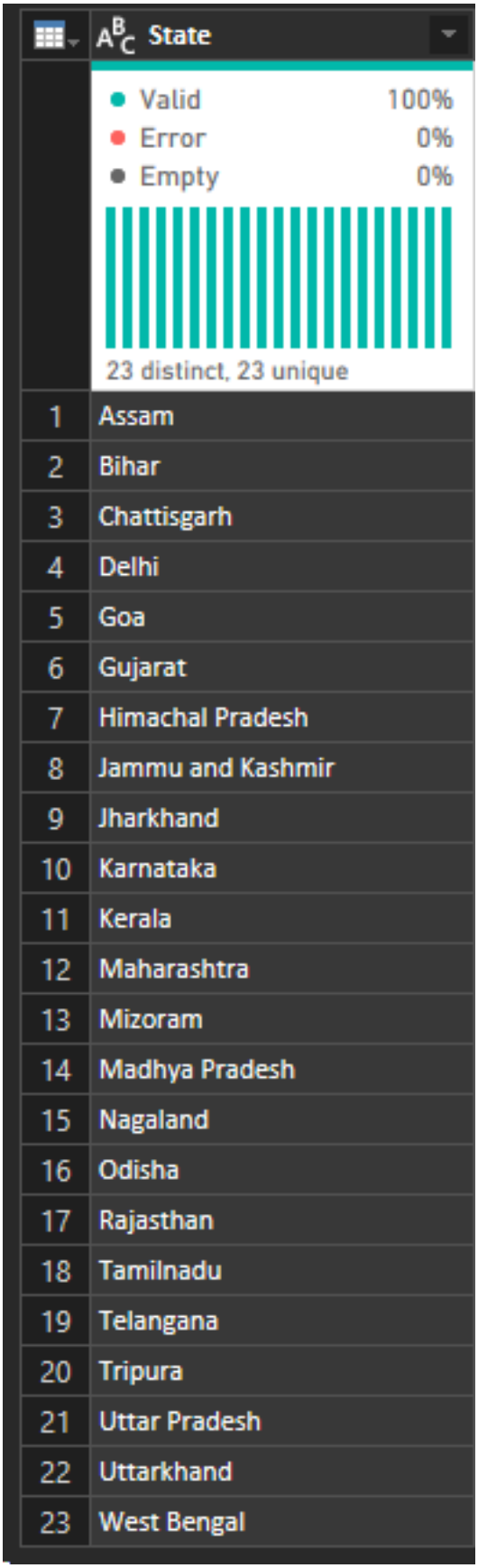
Used to:

- Analyze different types of posts

5.3.7 Dimension Table State

Used to:

- Compare engagement across locations



	State
1	Assam
2	Bihar
3	Chattisgarh
4	Delhi
5	Goa
6	Gujarat
7	Himachal Pradesh
8	Jammu and Kashmir
9	Jharkhand
10	Karnataka
11	Kerala
12	Maharashtra
13	Mizoram
14	Madhya Pradesh
15	Nagaland
16	Odisha
17	Rajasthan
18	Tamilnadu
19	Telangana
20	Tripura
21	Uttar Pradesh
22	Uttarkhand
23	West Bengal

Figure 30: Dimension Table State

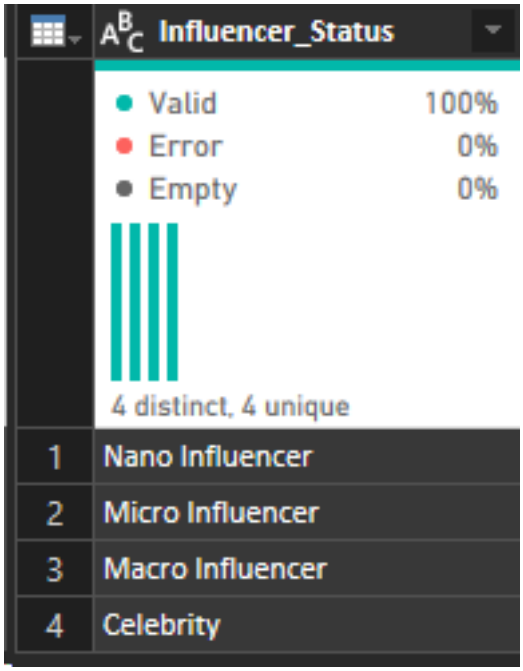


Figure 31: Dimension Table Influencer Status

5.3.8 Dimension Table Influencer Status

Used to:

- Compare high vs low influence accounts

5.3.9 Dimension Table Phase

Used to:

- Analyze how engagement changed over time

5.3.10 Dimension Table Verbosity

Used to:

- Write something here

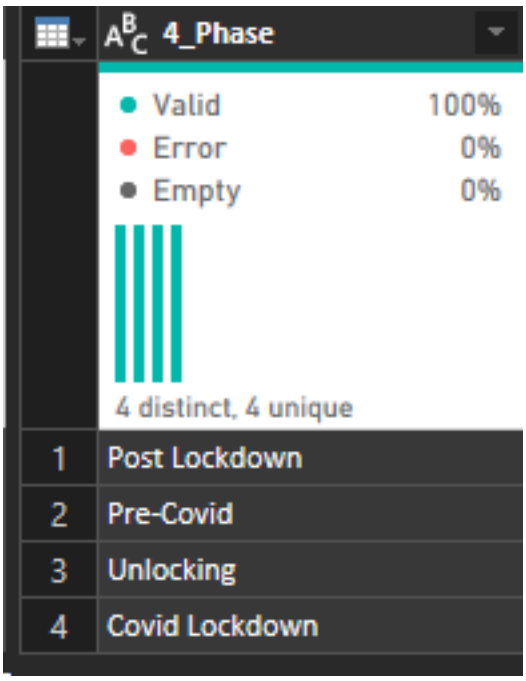


Figure 32: Dimension Table Phase

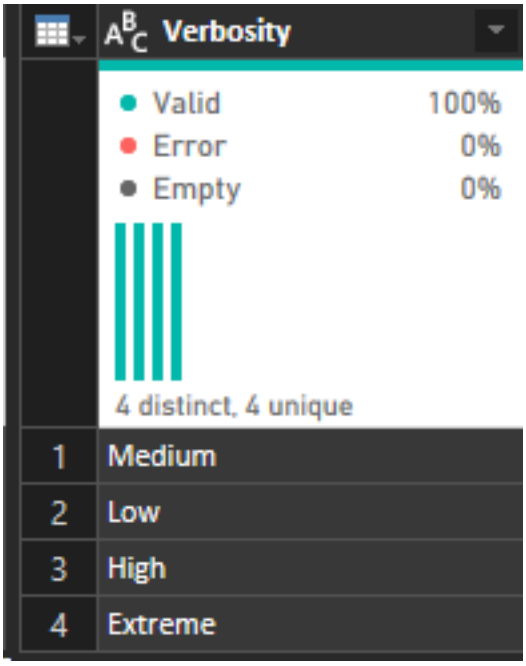


Figure 33: Dimension Table Verbosity

5.4 Relationships

All tables are connected in this way:

Dimension → Fact (One-to-Many relationship)

This means:

- One value in a dimension (e.g. one date)
- Can relate to many tweets in the fact table

5.4.1 Example:

- One date → many tweets
- One content type → many tweets
- One state → many tweets

5.5 Important Points

- All dimension tables connect only to the fact table
- Dimension tables do not connect to each other
- Filtering always goes from dimension → fact

5.5.1 Example of how it works

If you select:

- Content Type = Video
- Day Type = Weekend

The model will filter the fact table and show:

- Only weekend video tweets
- Then calculate engagement

5.6 Summary

- The fact table stores numbers (engagement, likes, sentiment)
- Dimension tables store descriptions (date, content type, state, etc.)
- The model allows you to analyze data from many perspectives easily

6 DAX Measures

6.1 Overview

We developed a set of DAX measures to support business questions. Measures are organized by functional category and stored in a dedicated _Measures table for clean model organization.

6.2 Complete Measures List

#	Measure Name	Category	Description
1	Total Posts	Aggregation	Count of all published posts
2	Total Likes	Aggregation	Sum of all likes received
3	Total Retweets	Aggregation	Sum of all retweets/shares
4	Total Engagement	Aggregation	Sum of all engagement actions (likes + retweets + replies + quotes)
5	Avg Engagement per Post	Aggregation	Average engagement per post

#	Measure Name	Category	Description
6	Avg Engagement	Aggregation	General average engagement
7	Avg Positive Sentiment	Aggregation	Average positive sentiment score
8	Avg Negative Sentiment	Aggregation	Average negative sentiment score
9	Engagement MTD	Time Intelligence	Month-to-date total engagement
10	Engagement YoY % Change	Percentage Change	Year-over-year growth rate
11	Engagement QoQ % Change	Percentage Change	Quarter-over-quarter growth rate
12	Engagement Rate	KPI	Engagement relative to audience size
13	Engagement Growth Trend	KPI	Change in growth rate (accelera-tion/deceleration)
14	Net Sentiment Score	KPI	Positive minus negative sentiment balance
15	State Engagement Rank	Ranking	State ranking by total engagement (1 = highest)
16	Post Type Share of Engagement	Advanced	Percentage of total engagement by post type
17	Top Post Type	Advanced	Highest-performing content format

#	Measure Name	Category	Description
18	High Positive Sentiment Engagement	Advanced	Avg engagement for posts with above-average positive sentiment
19	Low Positive Sentiment Engagement	Advanced	Avg engagement for posts with below-average positive sentiment

7 Complex Measure Explanations

Three advanced measures are highlighted below to demonstrate DAX proficiency and business logic implementation:

7.1 1. Top Post Type

Business Question: Which content format drives the most engagement?

What it does:
Identifies the single highest-performing content format (image, video, link, or text) based on total engagement.

How it works:
Creates a virtual table grouping posts by type with their total engagement, selects the top performer, and returns it as readable text.

Functions Used: `VAR`, `TOPN`, `SUMMARIZE`, `CONCATENATEX`

Key Techniques:
Unlike a simple `MAXX` which only returns a number, this measure returns the actual category name (e.g., "Video") in a stakeholder-friendly format.

Business Value:

Directly answers "What type of content should we create more of?" Enables data-driven content strategy and resource allocation.

7.2 2. Post Type Share of Engagement

Business Question: What percentage of total engagement does each content format contribute?

What it does:

Calculates the percentage contribution of each content type to total engagement.

How it works:

First calculates total engagement across all post types (ignoring any filters), then divides the current post type's engagement by that total.

Functions Used: `CALCULATE`, `ALL`, `DIVIDE`

Key Techniques:

Demonstrates the essential `CALCULATE` + `ALL` pattern required for percentage-of-total calculations.

Business Value:

Reveals engagement concentration—for example, if videos drive 60% of engagement but only represent 20% of posts. Helps balance content strategy.

7.3 3. Low Positive Sentiment Engagement

Business Question: Does below-average positive sentiment hurt engagement performance?

What it does:

Measures average engagement specifically for posts with below-average positive sentiment.

How it works:

Dynamically determines the average positive sentiment score, filters the fact table to include only posts scoring at or below that threshold, then calculates average engagement.

Functions Used: `CALCULATE`, `FILTER`

Key Techniques:

Pairs with High Positive Sentiment Engagement to create a true A/B test comparing performance across sentiment groups.

Business Value:

Answers the critical question "Does positive content actually drive higher engagement?" Quantifies ROI of maintaining positive brand tone.

8 Design Strategy

8.1 Radar Chart — Engagement across COVID phases by influencer type

A radar chart was chosen because it allows simultaneous comparison of multiple categories (COVID phases) across multiple groups (influencer tiers) in a single view. The concentric shape makes it immediately obvious which influencer tier covers the most "area" — visually communicating dominance without needing to read every number. A bar chart would have required multiple panels to achieve the same comparison.

8.2 Clustered Bar Chart — Verbosity vs engagement by influencer type

A clustered bar chart was chosen because it allows direct side-by-side comparison of engagement levels across verbosity categories for each influencer tier. Grouping bars by verbosity level makes the trend within each tier immediately readable. A line chart could have worked but would imply continuity between verbosity levels which are categorical, not continuous.

3a & 3b. Stacked Bar Chart — Sentiment expression and reception A diverging stacked bar chart was chosen because sentiment data is inherently bipolar — positive on one side, negative on the other. Anchoring both bars at zero makes it visually intuitive to compare the magnitude of positive versus negative sentiment simultaneously. Splitting into two charts (Information vs Interaction) was necessary to avoid conflating what influencers express with what audiences direct back at them, addressing the core analytical distinction your group member flagged.

8.3 Clustered Bar Chart — Followers by state

A clustered bar chart was chosen because it allows comparison of follower counts across multiple states while simultaneously showing the breakdown by influencer tier within each state. This makes it immediately visible that high follower counts in certain states are driven exclusively by Celebrity influencers rather than a broad mix of tiers.

8.4 Stacked Bar Chart — Post type distribution by influencer tier

A stacked bar chart was chosen because it shows both the total posting volume per influencer tier and the composition of post types within that total simultaneously. This dual function — volume and proportion in one view — makes it the most efficient chart for this business question. The corrected version uses count of posts rather than count of post types, ensuring the segments reflect actual posting behaviour and volume differences.

8.5 Stacked Column Chart — Post type vs engagement

A stacked column chart was chosen to show how total engagement is distributed across post types for each influencer tier. The vertical orientation naturally suits comparison of engagement magnitude, with taller bars immediately signalling higher performing tiers. Splitting by content type (Interaction vs Information) within each stack adds a second dimension without requiring a separate chart.

8.6 Clustered Column Chart — Verbosity vs engagement by content type

A clustered column chart was chosen over a stacked version because the goal here is to compare Interaction and Information content directly at each verbosity level, not to show their combined total. Side-by-side columns make the crossover points — where one content type overtakes the other — clearly visible at a glance.

8.7 Scatter Plot — Word count vs engagement

A scatter plot was chosen because it is the only chart type that can reveal correlation between two continuous variables — word count and engagement — across thousands of individual posts simultaneously. Each dot represents one post, allowing patterns, clusters, and outliers to emerge naturally. Trend lines per influencer tier were added to surface directional relationships that would be invisible in an aggregated bar chart.

8.8 Map Visual — Regional engagement

A map visual was chosen because the business question is inherently geographic — asking which states perform highest. Plotting on an actual map allows spatial patterns and regional clusters to emerge in a way that a bar chart ranked by state name never could. It also makes the report more accessible to stakeholders who think in terms of geography rather than data tables.

8.9 Box Plot — Engagement distribution across COVID phases

A box plot was chosen because the business question is about distribution and variability, not just averages. Showing the median, interquartile range, and whiskers for each COVID phase reveals how consistently or inconsistently posts performed within each phase — something a simple bar chart of averages would completely obscure. The V-curve pattern of engagement volatility only becomes visible when the full distribution is shown.

8.10 Heatmap Matrix — Engagement by day and hour across phases

A heatmap was chosen because it needs to display three dimensions simultaneously — day, time period, and COVID phase — in a compact, readable format. Colour intensity as the encoding mechanism allows the eye to immediately identify hotspots and cold zones without reading individual numbers. No other single chart type could display this three-way combination as clearly.

8.11 Line Chart — Engagement over time

A line chart was chosen because time-series data is continuous and directional — connecting data points with lines communicates the flow and trajectory of engagement over time naturally. Multiple lines per influencer tier allow trend comparison across the full timeline. A bar chart would fragment the time dimension and make long-term trends harder to follow visually.

9 Business Insights

9.1 How has Engagement Grown over Time by Year and Month?

This line chart tracks the Engagement Growth Trend across 2020–2022. It plots the period-over-period change rate, revealing where growth accelerated or declined.

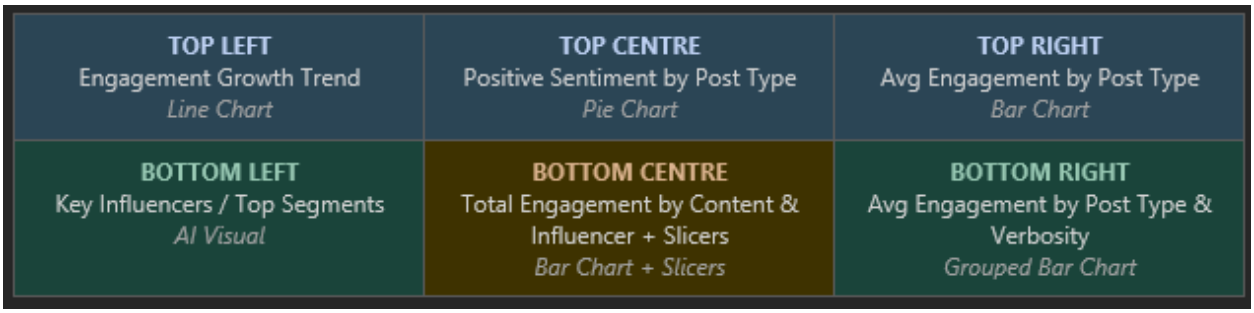


Figure 34: alt text

9.1.1 Findings

2020 Volatility: The chart shows extreme swings early in 2020, with growth spiking sharply positive before dropping to its most negative point (approximately –4), consistent with the disruption of COVID lockdowns creating an initial content surge followed by audience fatigue.

2021 Stabilisation: Growth rates moderate significantly through 2021, oscillating in a tighter band around zero. This reflects audiences re-establishing regular consumption patterns as restrictions eased.

2022 Plateau: By 2022, the growth trend flattens and hugs the zero line, indicating that engagement levels have stabilised rather than growing. The market appears to have reached a post-pandemic equilibrium.

Negative Dip (Mid-2020): The most dramatic negative trough, around –4, aligns with peak lock-down fatigue. Even as audiences were home and online, overall engagement growth collapsed, mirroring findings from the box plot analysis of COVID phases.

9.2 How is Positive Sentiment Distributed across Post Types?

This pie chart breaks down the total sum of Positive Sentiment by Post Type, showing which content formats collectively generate the most positive audience response.

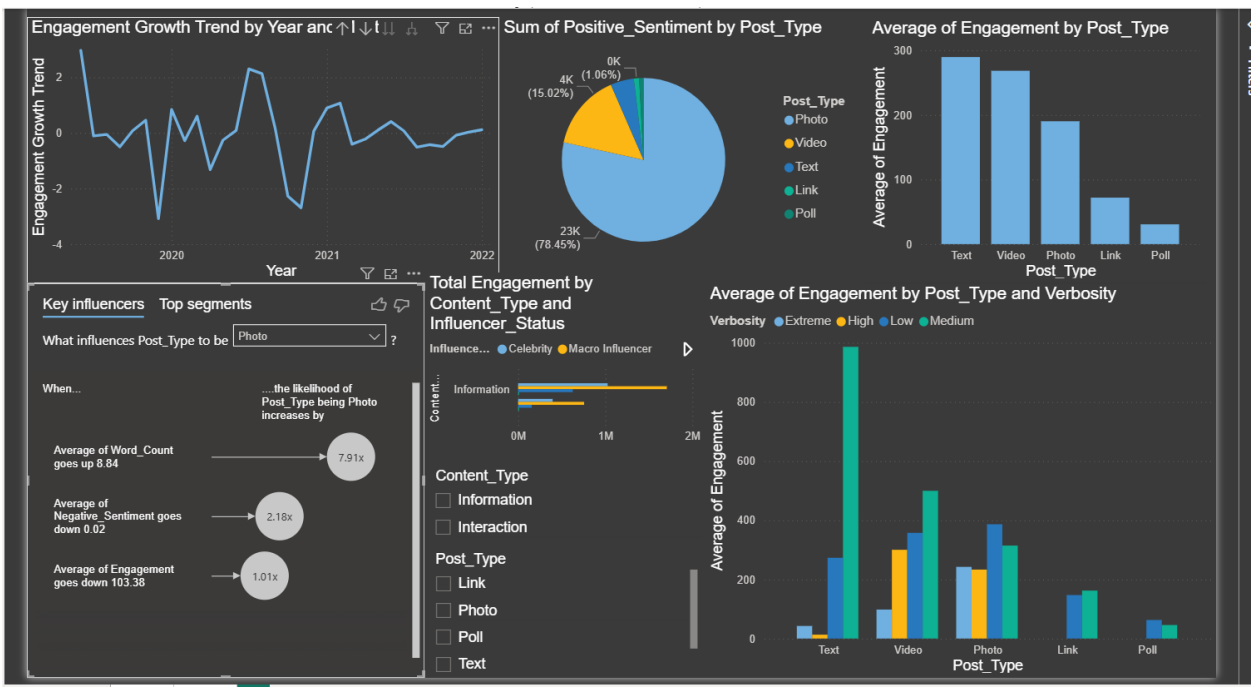


Figure 35: alt text

9.2.1 Findings

Photo dominates overwhelmingly: With 78.45% (23K) of all positive sentiment, Photo posts are by far the most positively received content format. This aligns with the finding that photo is also the most frequently posted type across all influencer tiers.

Video holds a meaningful share: Video accounts for a notable secondary slice, confirming its elevated engagement performance observed in the stacked column chart analysis. Audiences respond positively even though video is rarely used.

Text posts contribute modestly: Text represents 15.02% (4K) of positive sentiment, respectable given it is the second most commonly used post format, though far behind photo.

Link and Poll are negligible: Together, Link (4.62%, 1K) and Poll account for a very small fraction of positive sentiment, confirming that these formats rarely resonate with audiences across the dataset.

9.3 Which Post Type Generates the Highest Engagement?

This bar chart compares the Average of Engagement across five post types: Text, Video, Photo, Link, and Poll; giving a fair, volume-neutral comparison of which format resonates most per post.

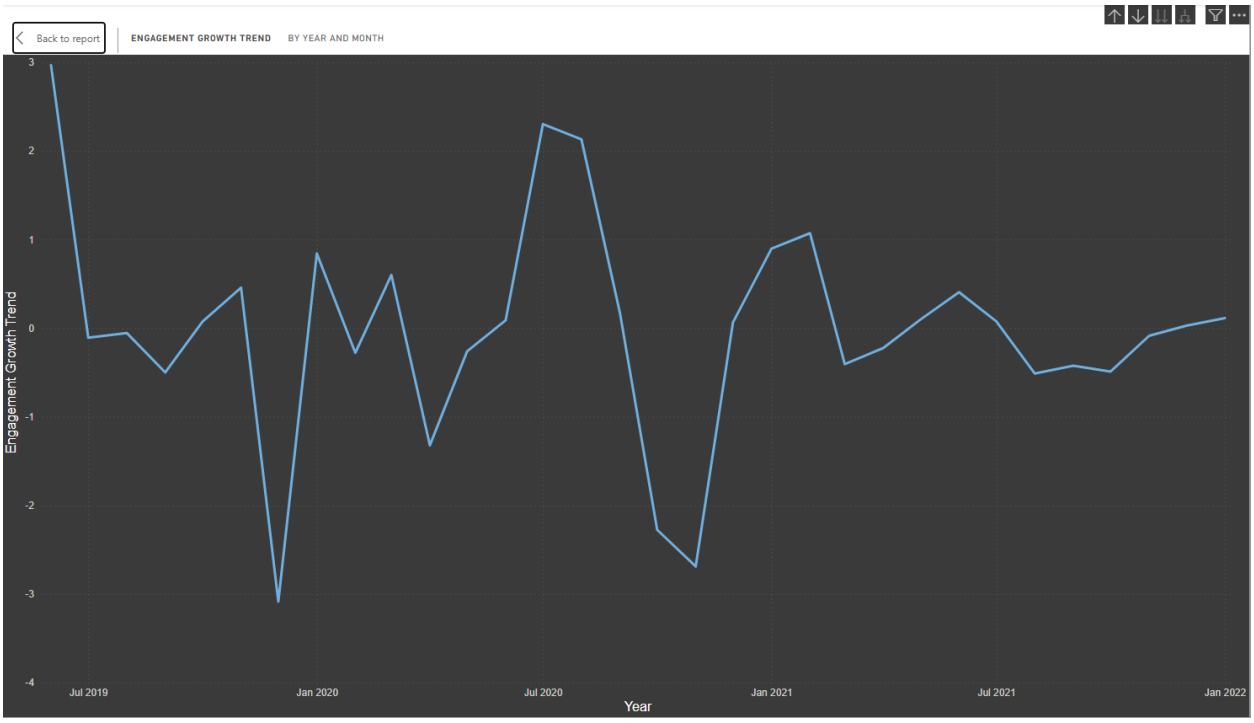


Figure 36: alt text

9.3.1 Findings

Text leads on average: Counterintuitively, Text posts register the highest average engagement (approximately 285), edging out Video in this view. This may be driven by a small number of high-performing text posts inflating the average, or by the fact that text posts from high-tier influencers skew the mean.

Video is a close second: Video averages approximately 265 engagements, confirming its strong performance. Combined with the stacked column findings (where Video has the highest engagement per tier), Video remains a premium-performing format.

Photo performs well in volume but modestly in average: Despite dominating total sentiment and posting volume, Photo's average engagement (~190) sits below both Text and Video, suggesting diminishing returns as photo volume increases.

Link performs below expectations: With an average around 65, Link posts generate relatively low engagement, suggesting audiences on this platform are reluctant to leave for external content.

Poll barely registers: Poll has the lowest average engagement of all post types, confirming it as the least effective format in this dataset.

9.4 What Factors most Influence the Likelihood of a Post being Classified as a Photo?

This Power BI Key Influencers visual identifies the conditions under which a post is most likely to be a Photo post, quantifying each factor's multiplicative impact on that likelihood.

9.4.1 Findings

Word Count (7.91×): When the Average Word Count goes up by 8.84 words, the likelihood of the post being a Photo increases by 7.91 times. This is the single strongest driver. It may initially seem counterintuitive — one might expect photos to accompany short captions — but it suggests that photo posts in this dataset tend to carry longer descriptive captions or hashtag-heavy text.

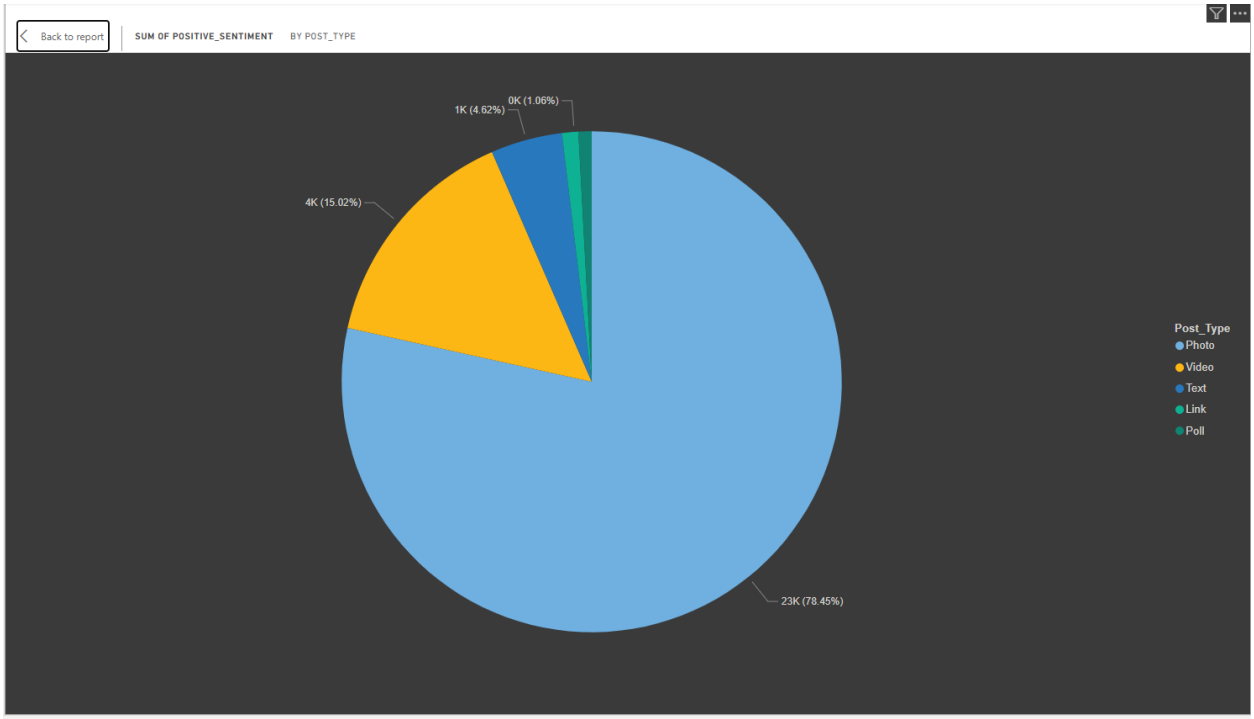


Figure 37: alt text

Negative Sentiment (2.18×): When Average Negative Sentiment decreases by 0.02, the likelihood of a Photo post increases by 2.18 times. Lower negativity is associated with photo content, reinforcing the idea that photo posts tend to be more positive and polished in tone.

Engagement (1.01×): When Average Engagement decreases by 103.38, the likelihood of a Photo post increases marginally (1.01×). This is a near-neutral effect — engagement alone is not a meaningful predictor of photo format selection.

9.5 How does Total Engagement compare Across Content Types for Different Influencer Tiers?

This chart compares Total Engagement for Information and Interaction content types across different influencer groups.

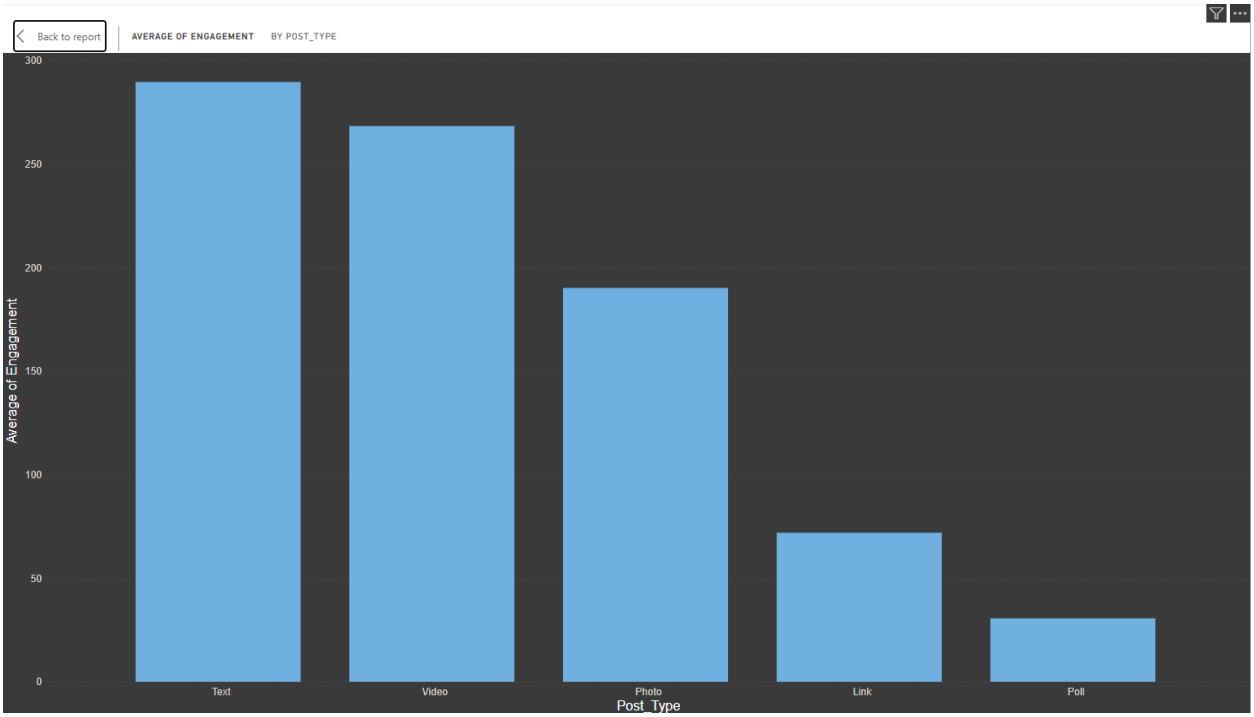


Figure 38: alt text

9.5.1 Findings

Interaction content leads for both groups: Both Celebrity and Macro Influencer bars are longer for Interaction content than for Information content. This universally confirms that audience engagement is higher when content is interactive rather than purely informational — consistent with the verbosity and stacked column analyses.

Celebrities generate more total engagement in Interaction: Celebrity Interaction bars extend noticeably further (approaching ~2M total engagements) compared to Macro Influencer Interaction, reinforcing the tier-based engagement hierarchy.

Information content is more evenly matched: The gap between Celebrities and Macro Influencers narrows for Information content, suggesting that the celebrity advantage is more pronounced for interactive posts.

Celebrity advantage is format-dependent: Celebrities do not simply outperform across the board, their lead is most pronounced in Interaction content, suggesting their audiences engage most when the content invites a response rather than delivers information.

How does Verbosity Interact with Post Type to affect Engagement? This grouped bar chart cross-analyses Average Engagement by both Post Type and Verbosity level (Extreme, High, Low, Medium), revealing whether the optimal word count varies by the type of post being created.

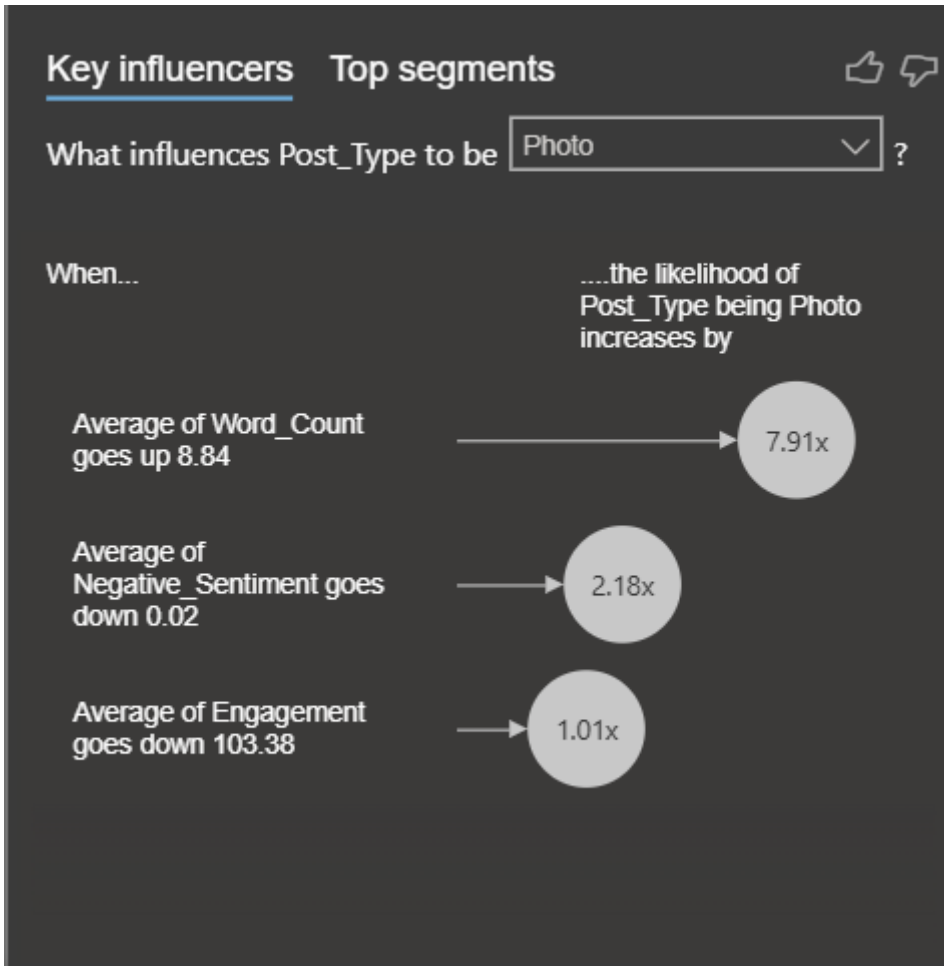


Figure 39: alt text

9.5.2 Findings

Text at Extreme verbosity peaks dramatically: The most striking finding is that Text posts at Extreme verbosity (purple bar) reach nearly 1,000 average engagement — far above any other combination in the chart. This is an outlier result suggesting that long-form text content, when it does break through, generates outsized engagement.

Video performs consistently across verbosity levels: Video posts maintain strong average engage-

ment across all verbosity levels, with the Medium and Low categories performing particularly well (~300–500). This reinforces video's position as the most reliable high-performing post format.

Photo shows moderate, stable performance: Photo posts deliver consistent but moderate engagement across all verbosity levels, with no single verbosity category dramatically outperforming others. Photo remains a volume format rather than an engagement spike driver.

Link and Poll remain low across all verbosity levels: Regardless of how much text accompanies them, Link and Poll posts generate very low engagement. Verbosity cannot rescue underperforming formats.

Extreme verbosity is high-risk, high-reward: The extreme verbosity category shows the widest spread of results — from very high (Text) to very low (Link/Poll). Short to medium captions appear safer for most formats.

9.6 What Post Type do Different Segment Influencers Post?

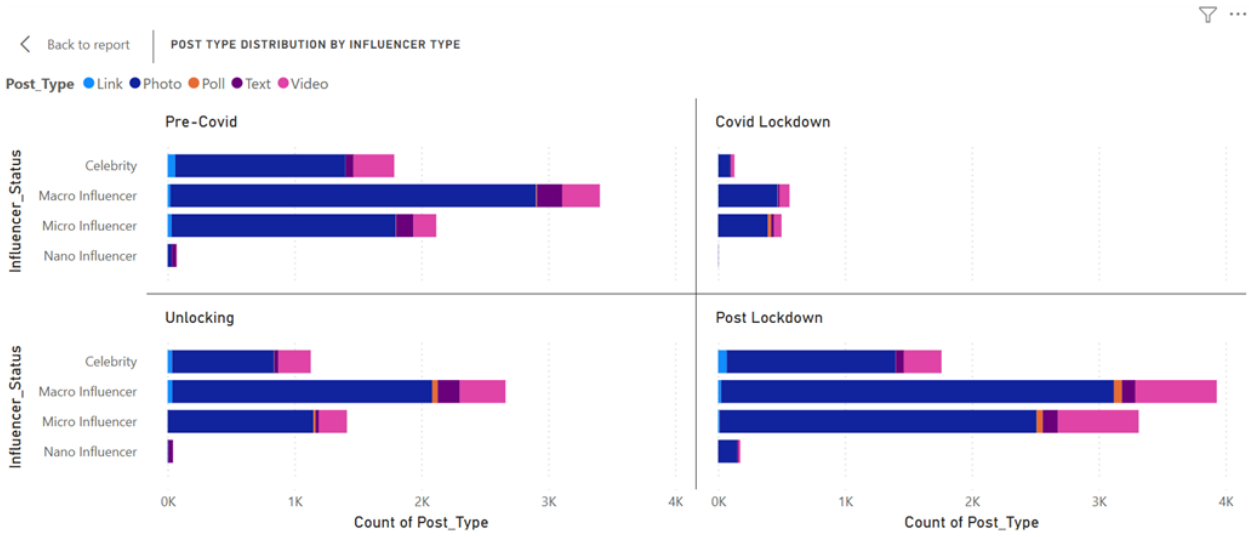


Figure 40: alt text

9.6.1 Findings

Photos dominate: Photos are the main content type for all influencer tiers at all times.

Macro influencers lead: They post the most, peaking at nearly 4,000 posts after lockdown.

Nano influencers lag: They post the least, rarely going above 200 posts.

Pandemic drop: Posting decreased significantly during lockdown for all tiers.

Strong recovery: Activity increased after lockdown, with Macro and Micro influencers surpassing pre-Covid levels.

Micro vs Celebrity: Micro influencers post more than Celebrities, especially before and after lockdown.

9.7 Which States have Socialites the Highest Number of Followers?

For this chart, we look at which regions drive the highest follower count; three states stand out: Kerala, Gujarat, and Odisha. Insights: What makes these states unique in the data is that their high average follower counts are driven exclusively by Celebrity influencers. We don't see a mix of smaller tiers reaching these numbers in these regions.

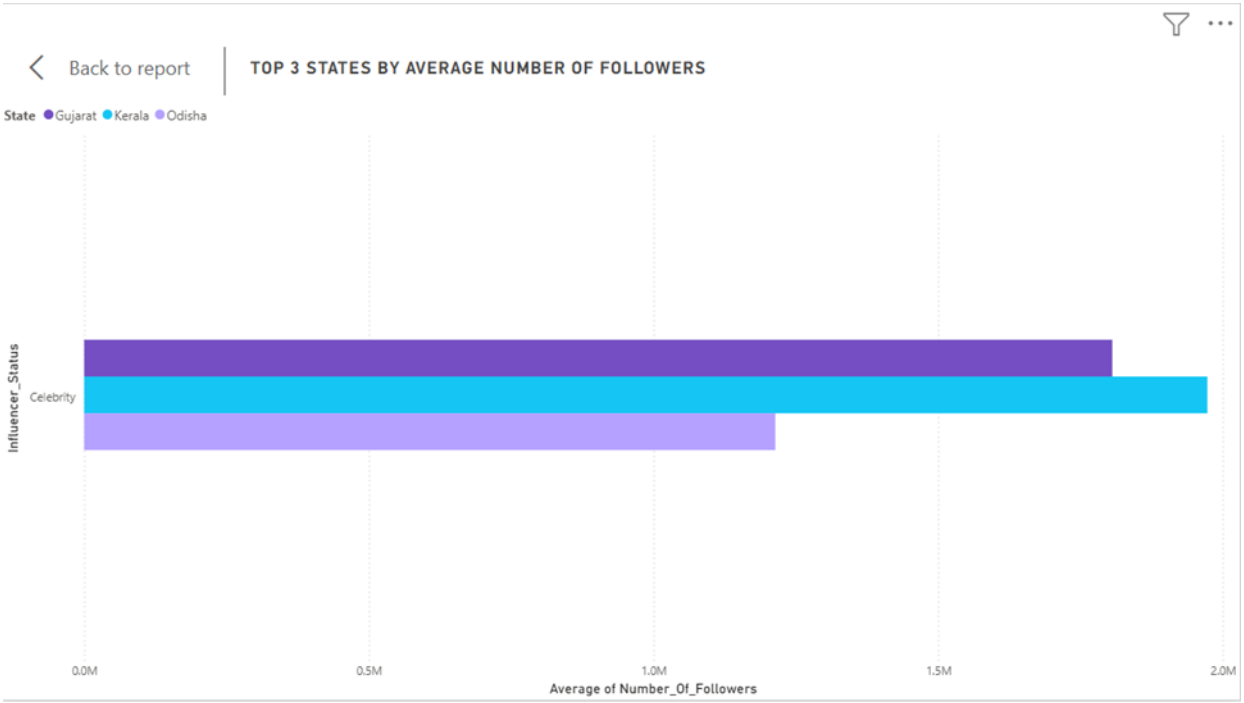


Figure 41: alt text

9.7.1 Findings

- Massive follower reach in these specific states is highly dependent on “Celebrity” status rather than a broad base of smaller creators.
- High-follower influence is not evenly distributed across the map; instead, it is geographically concentrated, with these three states acting as the primary hubs for the most followed influencers in our dataset.

9.8 How does Influencer Status affect Engagement across Phases?

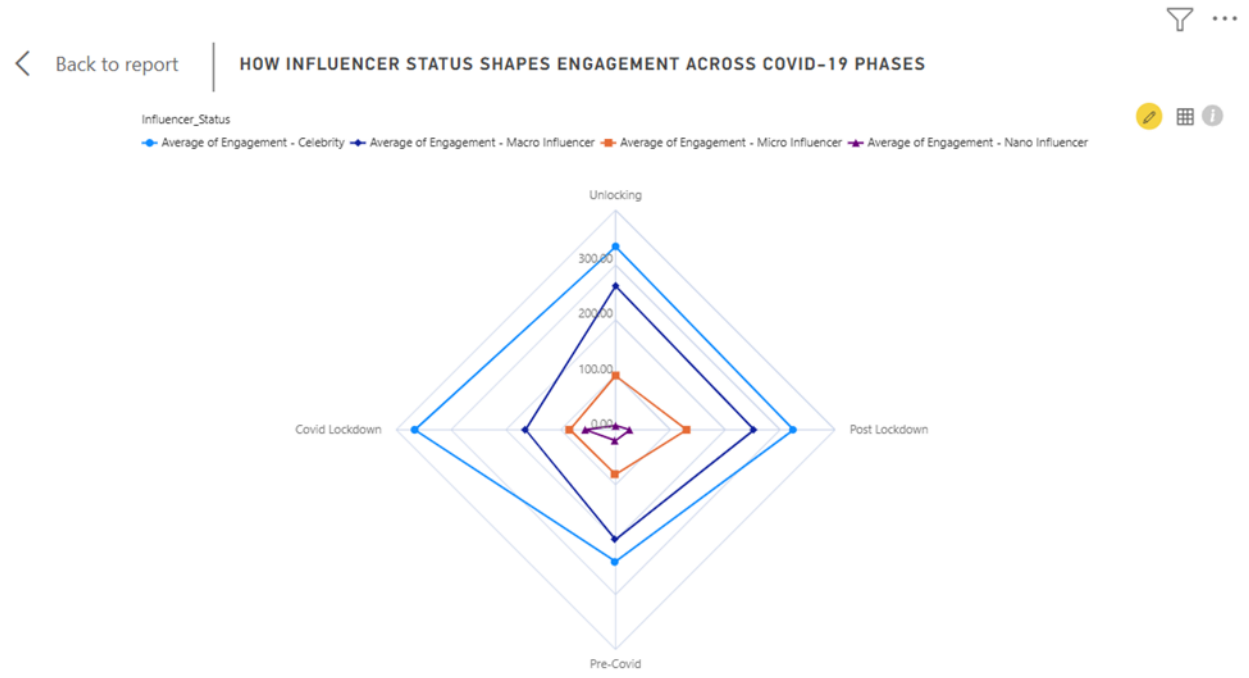


Figure 42: alt text

For this chart, we are looking at the engagement across the 4 COVID-19 phases and how it affects each influencer type.

9.8.1 Findings

Celebrities: We can see that they are in a league of their own. This is because they lead in every single phase. When everyone was stuck at home, their content was a go-to for most people. Their

lowest was pre-COVID, which is still higher than almost anyone else's peak.

Macro Influencers: They mirror the celebrities but on a smaller scale. Their performance peaked during the Unlocking phase, suggesting their content resonated most as people began transitioning back to normal life.

Micro Influencers: They saw their best results in the Post-Lockdown Period, which indicates that their engagement wasn't tied to a global crisis; instead, it grew as people settled back into their post-pandemic routines.

Nano-Influencer: They occupy the smallest area on the chart. While they saw a relatively high level during the COVID lockdown, their engagement dropped to its lowest point during the Unlocking phase.

9.9 How did Engagement Patterns Change across COVID Phases?

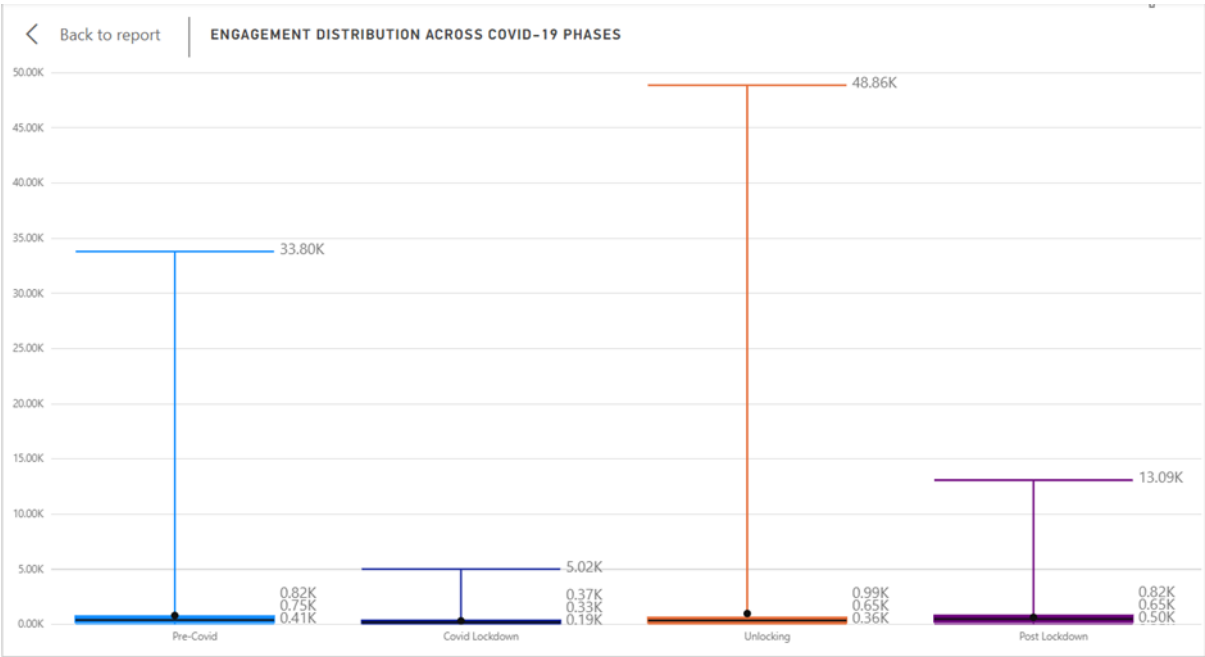


Figure 43: alt text

9.9.1 Findings

The "Pre-Covid" Baseline: Before the pandemic, engagement was naturally high, peaking at 33.80K.This represents our 'normal' world where people were consistently active and engaging with content at a high level.

The "Lockdown" Crash: During the Lockdown, engagement plummeted to its lowest point of 5.02K. When the lockdowns hit, engagement crashed. The pandemic effectively 'froze' activity, likely because people shifted their focus away from this specific type of content toward survival or essential needs.

The "Unlocking" Explosion: As soon as restrictions lifted, engagement hit an all-time record of 48.86K.This is the most interesting part. Once the world started 'Unlocking,' we saw a massive 'Rebound Effect.' The levels were even higher than before the pandemic as people rushed back to their old habits.

The "Post-Lockdown" Reality: Currently, engagement has settled at 13.09K, which is much lower than the original Pre-Covid levels. Finally, we see the 'New Normal.' Even though the surge was huge, it didn't last.People have adjusted to a new reality where engagement is much lower than before, suggesting that the pandemic caused a lasting shift in how people interact with content.

9.10 How do Engagement Levels Change over Time?

9.10.1 Findings

Pre-Covid Phase (Mar 2019 – Mar 2020): Engagement started relatively low but showed an early spike around Jul 2019 for Celebrities (~400), suggesting an active pre-pandemic digital environment. All influencer tiers maintained steady but modest engagement levels during this period.

Covid Lockdown (Mar 2020 – Jun 2020): Celebrity engagement spiked dramatically to ~575 around Jan 2020 just before lockdown, then experienced the most volatile period in the entire timeline. The biggest spike in the entire dataset occurs around Jul 2020 (~725) — right at the start of Post Lockdown — confirming that lockdown created extraordinary engagement opportunities for Celebrities.

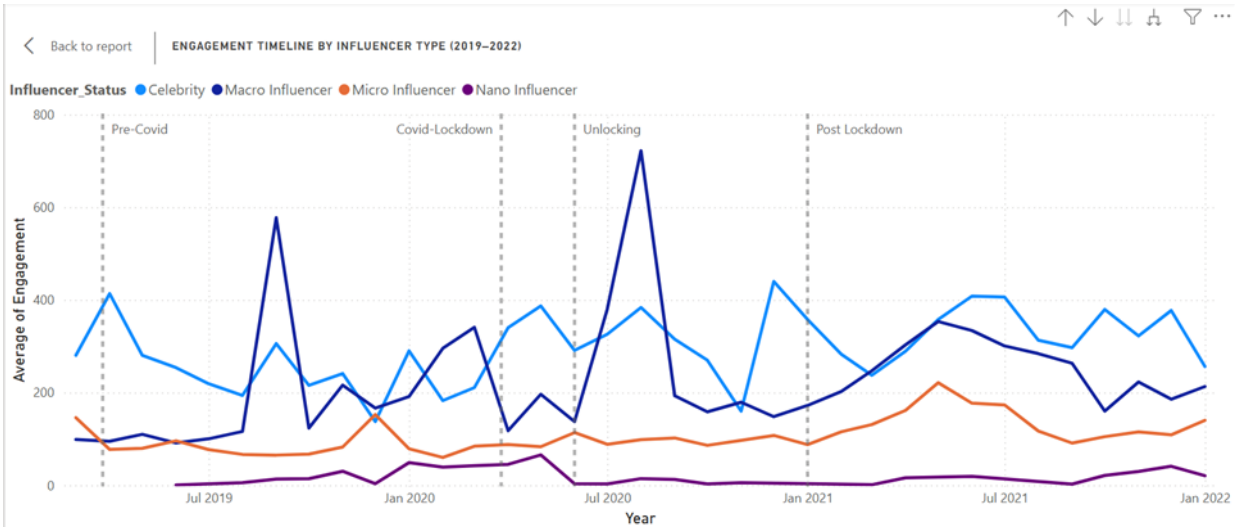


Figure 44: alt text

Unlocking (Jun 2020 – Jan 2021): Engagement remained elevated but began fluctuating significantly. Celebrities continued dominating while Macro Influencers showed a gradual upward trend, slowly closing the gap.

Post Lockdown (Jan 2021 – Jan 2022): Celebrity engagement gradually declined from its pandemic peak but stabilised at a higher level than pre-COVID. Macro Influencers continued their upward trajectory, reaching their highest levels during this phase. Macro Influencers show a clear upward trend, while Micro and Nano Influencers remain relatively flat, indicating that the pandemic's impact on engagement was most pronounced for higher-tier influencers.

9.11 Which Post Type Generates the Highest Engagement?

9.11.1 Findings

For celebrities:

- Video has the highest engagement
- Photo is second
- Interaction content outperforms Information across most post types

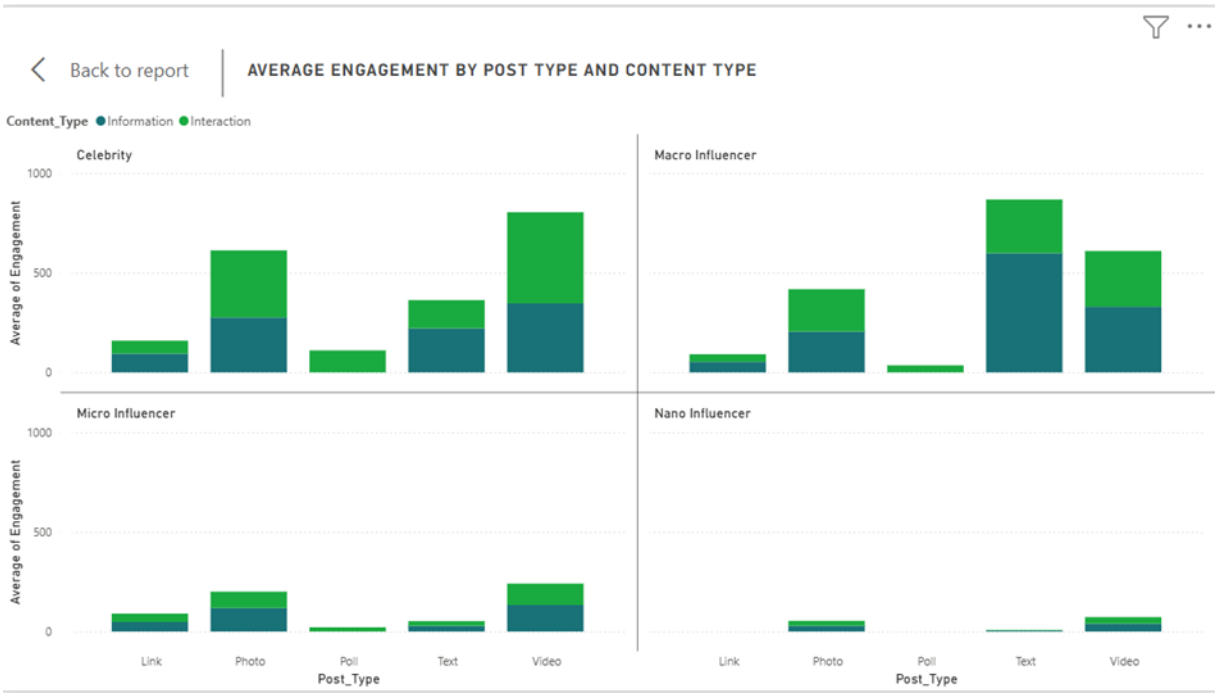


Figure 45: alt text

For macro influencers:

- Video also dominates
- Photo and Text are strong second options
- Similar pattern to Celebrities — Interaction content leads

For micro influencers:

- Video again leads
- Much lower overall compared to Celebrity and Macro
- Same Interaction > Information pattern holds

For nano influencers:

- Engagement is very low across all post types
- No single post type clearly dominates
- Barely registers compared to other tiers

9.12 Does Verbosity Impact Engagement Levels on Content Type?

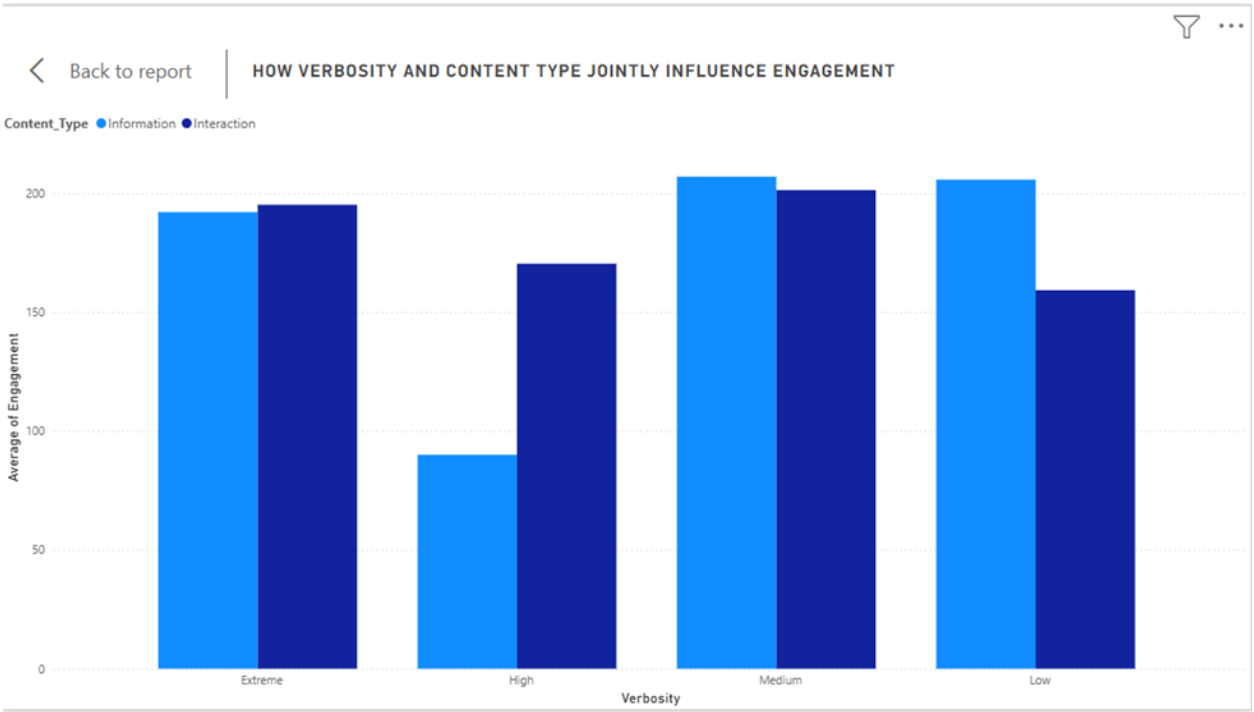


Figure 46: alt text

9.12.1 Findings

Medium verbosity drives the highest overall engagement regardless of content type

Interaction content is more resilient — it performs consistently well across all verbosity levels, never dropping as sharply as Information

Information content is most vulnerable at High verbosity — it drops significantly, suggesting audiences disengage from purely informational content when verbosity is high but not extreme

At Low verbosity, Information slightly outperforms Interaction — the only level where this reversal occurs

If you want consistent engagement, Interaction content is the safer bet across all verbosity levels. Information content is more volatile — it peaks at Medium and Low but struggles at High verbosity.

9.13 Does Verbosity affect Engagement Differently by Influencer Type?

For this chart, we look at how much influencer posts vs how much their audience interacts. We used Average Engagement scores to make our comparisons fair, regardless of how much each person posts.

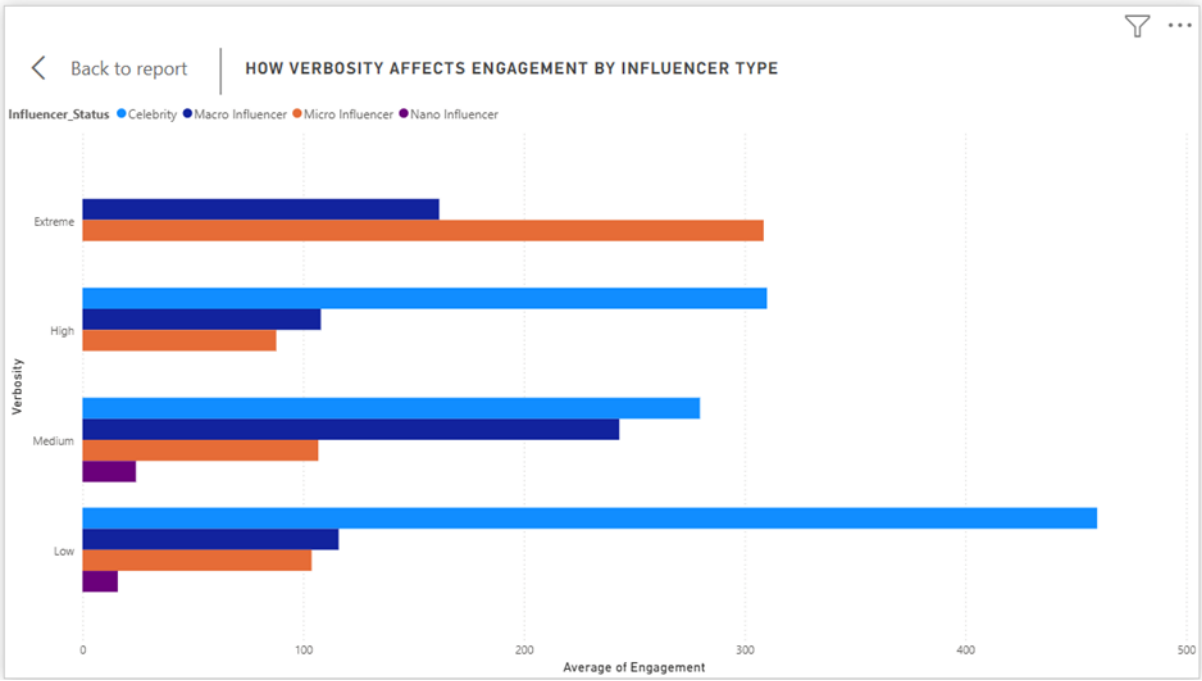


Figure 47: alt text

9.13.1 Findings

Celebrities: They don't need to say much on their social media, hence why their highest engagement is at "Low", hitting nearly 500 interactions. Interestingly, they completely disappear from the data entirely at the "Extreme" verbosity level, which suggests their audience prefers short, quick updates.

Micro-Influencers: They have to work harder, end up seeing an opposite trend where their engagement is low for short posts but takes the lead at "Extreme" verbosity.

Macro-Influencers: They have a middle ground where they perform consistently well but peak when they provide a moderate amount of content, nearly catching up to celebrities at the "Medium" level.

Nano-Influencer: Their engagement levels remain low across the board, showing very little change, no matter how much they write.

9.14 Does Word Count Influence Engagement Levels?

This scatter plot maps out every post by its word count versus the engagement it received. If writing more led to more likes, we would see the dots trending upward as we move to the right.

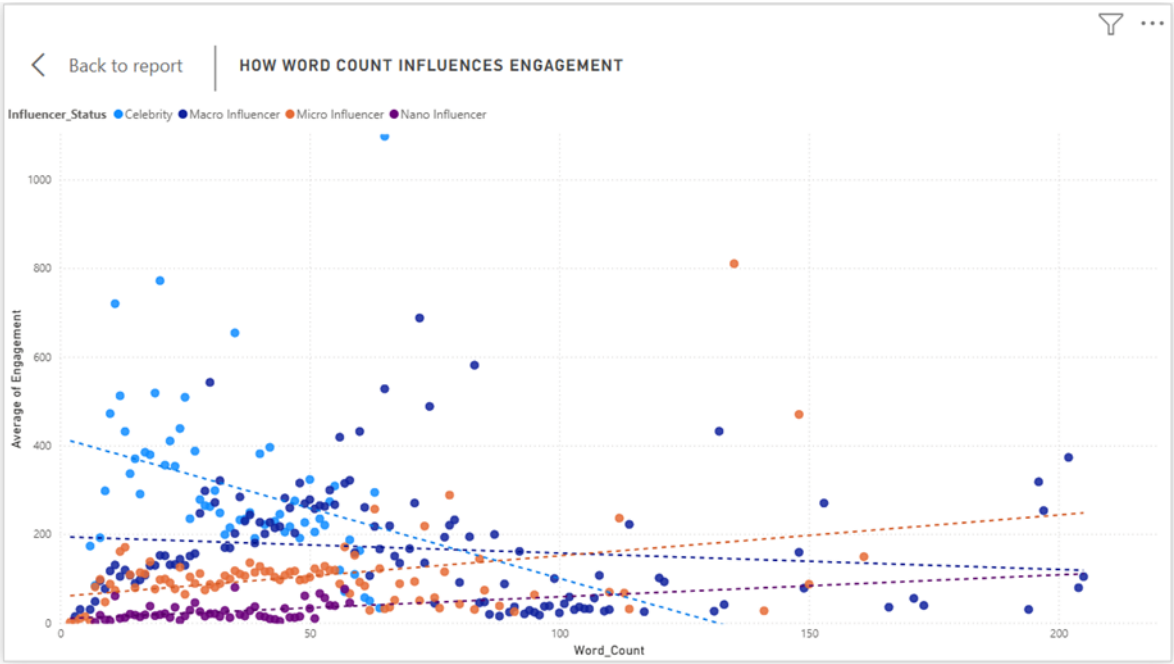


Figure 48: alt text

9.14.1 Findings

Celebrities "break" the chart at low word counts. The highest engagement spikes in the entire study happen when Celebrities say very little, usually under 50 words.

As we move toward the higher word counts (100–200 words), engagement stays remarkably flat for everyone else. Micro and Nano influencers stay grounded. Regardless of how much they write, these tiers cluster near the bottom of the engagement scale, showing that word count isn't a "ladder" they can use to climb into higher engagement brackets.

There is no meaningful correlation between word count and engagement. The data proves that who is posting is far more important than how much they write. For celebrities, fame acts as an amplifier for even the smallest amount of effort. For smaller influencers, writing longer posts doesn't compensate for a smaller reach.

9.15 Which States have the Highest Engagement?

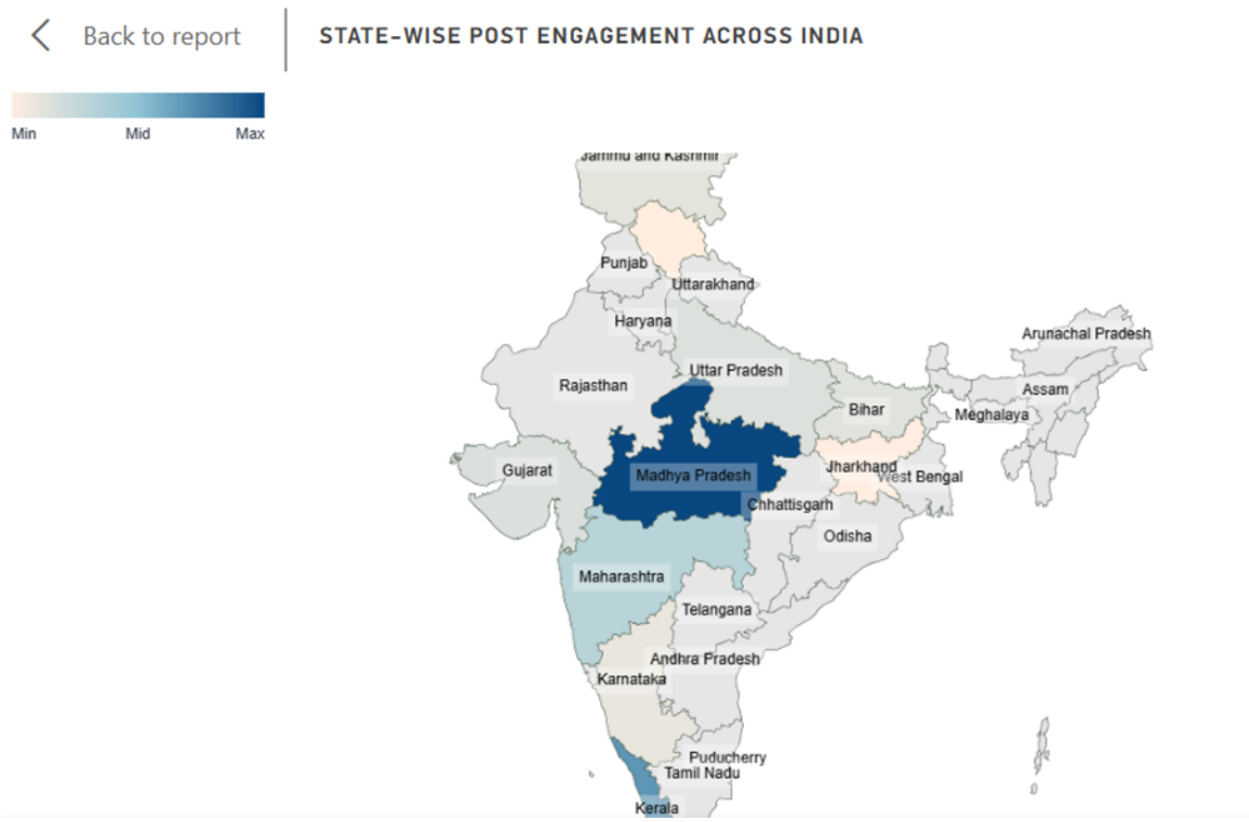


Figure 49: alt text

9.15.1 Findings

Maharashtra leads: It has the highest average engagement, driven by a strong presence of Celebrity influencers who generate significant interaction.

Tamil Nadu and Karnataka follow: These states also show high engagement levels, likely due to a combination of active influencer communities and engaged audiences.

Regional disparities: Engagement levels vary widely across states, suggesting that local factors such as population density, internet penetration, and cultural engagement with social media may play significant roles in driving interaction.

Celebrity influence: The states with the highest engagement are also those where Celebrity influencers are most active, reinforcing the idea that influencer presence is a key driver of engagement in these regions.

9.16 Which Day and Hour has the Highest Engagement across Phases?

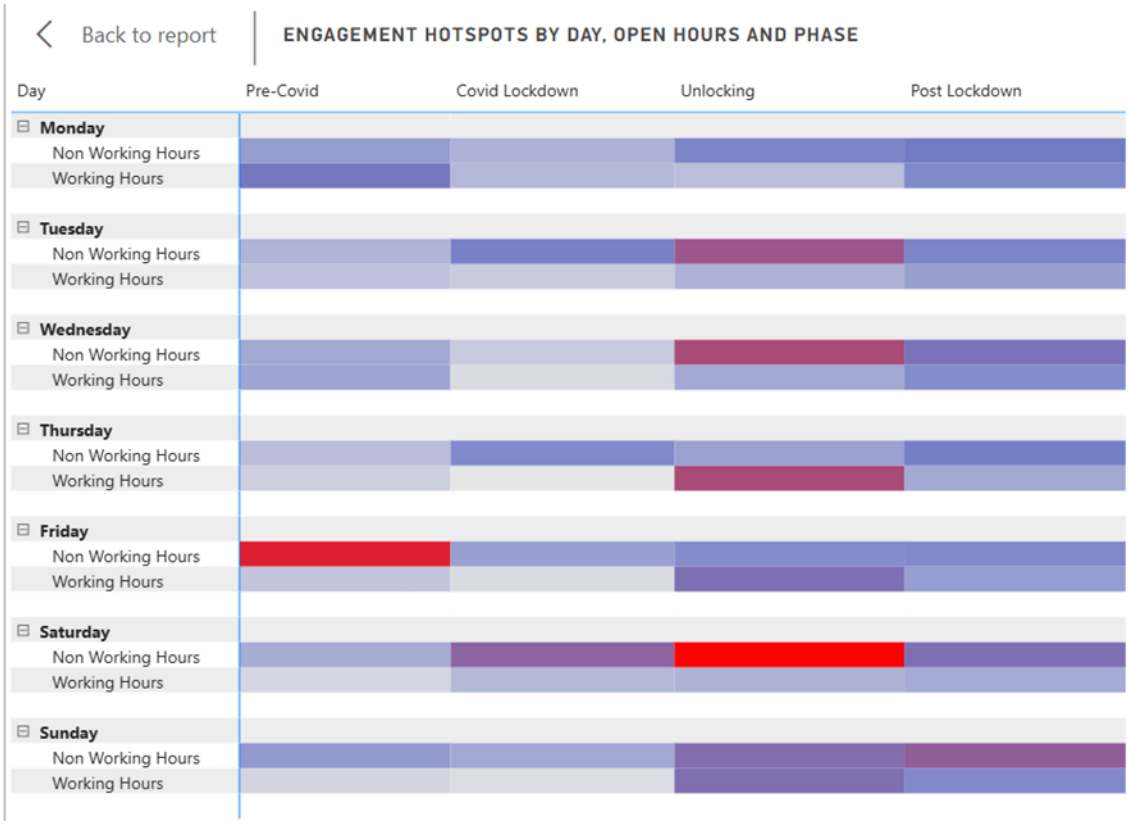


Figure 50: alt text

9.16.1 Findings

Overall Pattern: Non-Working Hours consistently outperform Working Hours across every day and every phase. This confirms that people engage with content most during their personal time rather

than work hours, regardless of the pandemic phase.

Friday Evening Peak: The single highest engagement hotspot is on Fridays during Non-Working Hours in the Pre-Covid phase (346.45), making it the most active time for engagement in the entire dataset. This suggests that people were most engaged with content at the end of the workweek before the pandemic disrupted normal routines.

Saturday Non-Working Hours: Unlocking is the second highest, suggesting weekend evenings during the return to normalcy were highly active.

Thursday Working Hours Unlocking (295.00) is surprisingly high for working hours, suggesting the Unlocking phase drove unusual weekday engagement.

Covid Lockdown Pattern: The Covid Lockdown column is consistently the lightest across all days and hours hence confirming our earlier finding that lockdown suppressed engagement across the board regardless of day or time.

Key Insight: Engagement follows human behaviour, people interact most with content during their personal time, not work hours. Friday evenings Pre-Covid represented the peak engagement window, while Covid Lockdown flattened all temporal patterns entirely, erasing the natural rhythm of online behaviour.

9.17 What Sentiments do Influencers Express in their Content?

For this chart, we were looking at what sentiment influencers actively express in their own content. The chart compares the average positive and negative sentiment scores across each influencer type, filtered to information-type posts only — that is, content created directly by the influencer.

9.17.1 Findings

Negative expression is identical across all tiers. Every influencer type posts with exactly −1.06. Tier, reach, and fame have zero influence on how negatively influencers frame their content.

Celebrities express the most positivity. At 1.35, Celebrities edge out all other tiers. Micro influencers express the least positivity at 1.24, with Macro and Nano tied at 1.25.

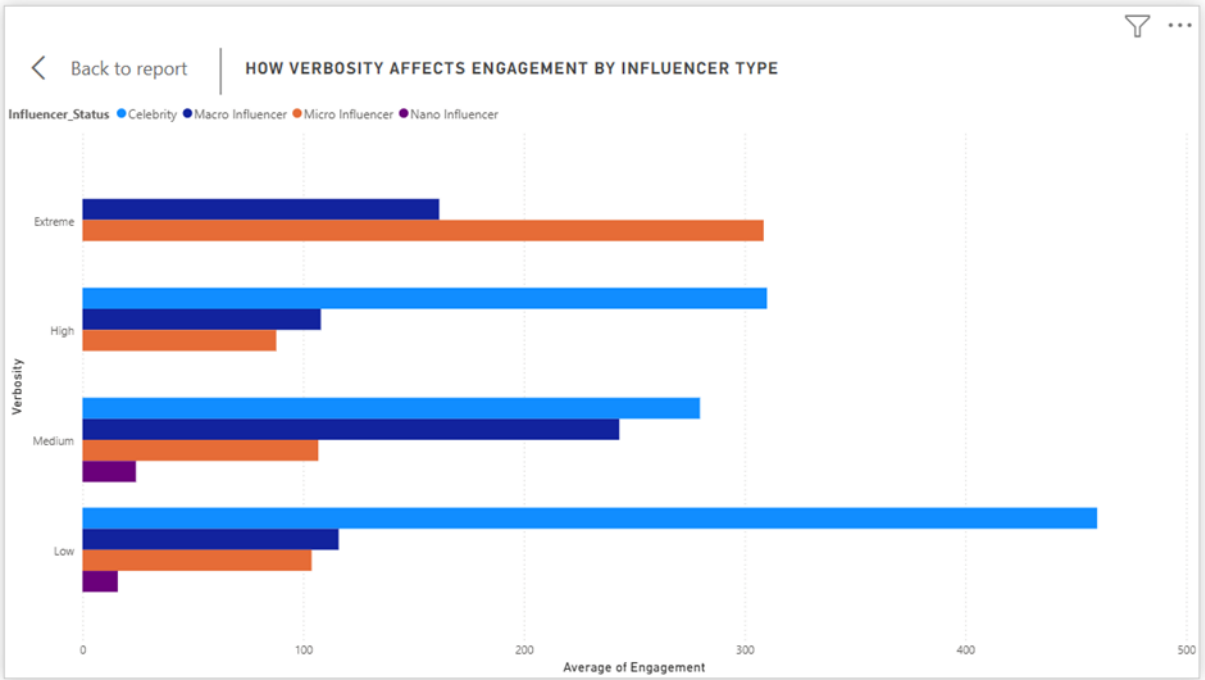


Figure 51: alt text

Positivity outweighs negativity for all tiers. Every tier sits net-positive, meaning that regardless of influencer status, content leans more positive than negative overall.

Influencer tier shapes how positively someone expresses themselves, but not how negatively. The negativity floor is universal — suggesting it reflects a general norm of content creation on the platform rather than individual influencer behaviour. Celebrities post the most positively, which may reflect more polished, brand-conscious content strategies rather than genuine differences in sentiment.

9.18 What Sentiment do Audiences Direct Back at Influencers?

For this chart, we examined the sentiment audiences directed at influencers through comments and interactions. The chart compares the average positive and negative sentiment scores across each influencer type, filtered to interaction-type posts only.

Figure 52: alt text

9.18.1 Findings

Negativity is mostly flat, but Nano influencers attract slightly more Celebrities and Micro influencers receive −1.06, while Macro influencers receive the least negativity at −1.04. Nano influencers stand out slightly at −1.09, the highest negative reception of all tiers, which is unexpected given their smaller reach.

Celebrities and Nano influencers receive the most positivity equally. Both sit at 1.38, jointly leading the audience's positive reception. Micro influencers follow at 1.28, while Macro influencers receive the least positivity at 1.22.

Macro influencers have the weakest audience reception overall. They receive the least positivity and the least negativity — suggesting their audiences are the most emotionally neutral or disengaged in their responses.

Unlike expression, reception does vary more noticeably across tiers. The most striking finding is that Nano influencers — despite the smallest reach — attract audience sentiment comparable to Celebrities. This suggests that community closeness at the Nano level drives stronger emotional reactions, both positive and negative, than the middle tiers

10 Final Delivery